

MI-algebras: A new framework for arithmetics of (extensional) fuzzy numbers

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This paper is a tribute dedicated to the memory of Milan Mareš

Abstract

The existing arithmetics of fuzzy numbers, usually stemming from the α -cut arithmetic, do not preserve some of the important properties of the standard arithmetics of real numbers. Although one cannot expect a generalization of standard arithmetic to perfectly preserve all its properties, the most important properties should be preserved at least in a weakened form. We present a novel framework for arithmetics of extensional fuzzy numbers that more or less preserves all the important (algebraic) properties of the arithmetic of real numbers and thus seems to be an important seed for further investigations on this topic. The suggested approach to arithmetics of extensional fuzzy numbers is demonstrated by many examples, and, besides its algebraic properties, it is also shown to carry some desirable practical properties. The investigation leads to novel algebraic structures – MI-algebras (MI-monoids, MI-groups, MI-fields) – that abstract the discussed properties. The main idea of these structures is based on a set of “pseudoidentities” that complements the only commonly used identity element in a monoid/group structure. These pseudoidentities are elements that generalize the identity-like behavior and allow us to weaken the standard forms of algebraic properties that are highly desirable to preserve at least in a weakened form. Appropriate properties of suggested MI-structures are introduced and demonstrated by a number of practical examples.

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1. Introduction

Because fuzzy numbers were designed to generalize standard real numbers, it is unquestionable that the arithmetics of fuzzy numbers should generalize standard arithmetics as well. The importance of fuzzy numbers, i.e., fuzzy sets representing vaguely defined real numbers, has been published many times [9,24]. Thus, developing working arithmetics of fuzzy numbers is obviously of similar importance. However, despite the tremendous work done in this field [9,11,12,24], to the best of our knowledge, so far the existing arithmetics of fuzzy numbers failed to maintain some very important algebraic properties that are naturally valid for the arithmetic of real numbers.

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By these very important algebraic properties, we basically mean the following three properties:

$$a + (-a) = 0, \quad (1)$$

$$a \cdot a^{-1} = 1, \quad (2)$$

$$(a + b) \cdot c = a \cdot c + b \cdot c, \quad (3)$$

i.e., the existence of an identity element for both summation and multiplication and the distributivity law, which are naturally valid for any $a, b, c \in \mathbb{R}$.

As we know, none of these properties (1)–(3) generally holds for $a, b, c \in \mathcal{F}(\mathbb{R})$ where $\mathcal{F}(\mathbb{R})$ denotes the set of all fuzzy sets on real numbers. And neither restriction to certain subsets of $\mathcal{F}(\mathbb{R})$ such as to LR-fuzzy numbers [32] or to LU-fuzzy numbers [41] helps sufficiently in order to obtain a kind of valid variant of the three properties recalled above.

In our opinion, the main problem is that fuzzy sets, although defined on the universe of real numbers and restricted by distinct technical conditions (e.g., upper semi-continuity of LR-fuzzy numbers), are still fuzzy sets. That is, they are generalizations of classical sets, not generalizations of real numbers that could be considered as vague representations of real numbers. Moreover, one can easily see that the classical (crisp) sets equipped with the arithmetical operations based on Zadeh's extension principle do not preserve the properties of the arithmetic of real numbers. Thus, we may hardly expect that their generalization to fuzzy sets would regain these lost properties. Let us recall that the fact that fuzzy numbers rather fuzzified version of crisp sets or more precisely intervals has been noted by several authors (see, e.g., [15,26,33]).

We present a novel approach to arithmetics of fuzzy numbers that combines several highly desirable basic principles such as the following:

- we restrict our considerations to fuzzy numbers that are conceptually created from real numbers because arithmetic operations work on numbers but not on crisp sets;
- we do not stick to Zadeh's extensional principle or the α -cut based arithmetic as the only basic principles from which our approach should stem;
- we capture an intuitive nature of arithmetics with vague numbers;
- and finally, we lower the required computational costs.

Remark 1.1. It should be noted that the above features and properties of arithmetics relate only to arithmetics that are α -cut based or that stem from Zadeh's extensional principle, but not to all existing arithmetics. For example, in set-valued analysis, we may find arithmetic operations defined in the Minkowski sense (see [1,2,17,29–31]). However, the above arithmetics that stem from α -cut calculus or from Zadeh's extensional principle are standard and most widely used, and therefore, they deserve our attention in this investigation. The same problems that we mention above may have also led other authors to focus on inappropriate behavior of the standard arithmetics of fuzzy numbers. Consequently, these authors published articles on constrained arithmetics that deserve attention for attempting to cope with the known problems (see [22,23,36]).

In addition to the implementation of the above listed principles, our main goal is to introduce a working (in a broad sense) scheme for arithmetics of fuzzy numbers that may be inherited for basically all purposes where fuzzy numbers are naturally used.

2. Preliminaries

2.1. Conceptual approach – basic notions

As foreshadowed above, we do not find all the fuzzy sets that are usually called fuzzy numbers to be appropriate candidates for treatment as summands or multipliers in generalized arithmetic operations. Once more, we stress that arithmetic operations do not generally work on crisp sets [15], and thus if fuzzy sets naturally extend crisp sets, one may hardly expect to obtain a working arithmetic on fuzzy sets. Therefore, the suggested arithmetic of fuzzy numbers

will be designed for fuzzy sets that are natural and mainly conceptual generalizations of real numbers, i.e., appropriate models of vaguely understood real numbers.

If a fuzzy set $a \in \mathcal{F}(\mathbb{R})$ is supposed to model the meaning, e.g., of an amount “about five”, it should naturally be constructed using the real number 5 and a tolerance of values close to 5. The closeness may be easily modeled within the fuzzy framework using the similarity relation [19] (also called fuzzy equivalence or fuzzy equality [20]) that generalizes the crisp equality. With respect to the goal of our paper, we restrict most of the definitions introduced in the latter to the universe of real numbers.

Definition 2.1. Let $*$ be a left-continuous t-norm. A binary fuzzy relation on the set of real numbers $S : \mathbb{R}^2 \rightarrow [0, 1]$ is called a $*$ -similarity if the following hold:

$$\begin{aligned} S(x, x) &= 1, \\ S(x, y) &= S(y, x), \\ S(x, y) * S(y, z) &\leq S(x, z), \end{aligned}$$

for all $x, y, z \in \mathbb{R}$.

Definition 2.2. Let $*$ be a left-continuous t-norm. A $*$ -similarity relation S on $\mathbb{R}$ is said to be *separated* if, for any $x, y \in \mathbb{R}$, it holds

$$S(x, y) = 1 \quad \Rightarrow \quad x = y,$$

and *shift-invariant* if, for all $x, y, h \in \mathbb{R}$, it holds

$$S(x, y) = S(x + h, y + h).$$

Remark 2.1. Note that it can be easily shown that the $*$ -similarity relation S is separated¹ if and only if

$$S(x, \cdot) = S(y, \cdot) \quad \Rightarrow \quad x = y \tag{4}$$

holds for any $x, y \in \mathbb{R}$.

The similarity relation generalizes the classical equality, and thus it is very useful in definitions of further notions that generalize the classical ones. For example, extensionality is a standard notion from the set theory that may be easily generalized for fuzzy sets with the help of the similarity relation (see, e.g., [6,7]).

Definition 2.3. (See [5].) Let S be a $*$ -similarity relation on $\mathbb{R}$. A fuzzy set $a \in \mathcal{F}(\mathbb{R})$ is said to be *extensional w.r.t. S* if

$$a(x) * S(x, y) \leq a(y)$$

holds for any $x, y \in \mathbb{R}$.

Extensional fuzzy sets compose an important class of fuzzy sets that represent collections of objects that have the same property if they are close to each other. Although not each fuzzy set is extensional, for each fuzzy set, we may define its extensional hull, which is an extensional fuzzy superset of the given fuzzy set.

Definition 2.4. Let S be a $*$ -similarity relation on $\mathbb{R}$, and let a be a fuzzy set on $\mathbb{R}$. A fuzzy set $\hat{a} \in \mathcal{F}(\mathbb{R})$ is said to be an *extensional hull of a w.r.t. S* if it is the least fuzzy superset of a that is extensional w.r.t. S .

Theorem 2.1. (See [19].) Let S be a $*$ -similarity relation on $\mathbb{R}$, and let a be a fuzzy set on $\mathbb{R}$. The extensional hull of a w.r.t. S is a fuzzy set $\hat{a} \in \mathcal{F}(\mathbb{R})$ given as follows

$$\hat{a}(x) = \bigvee_{y \in \mathbb{R}} a(y) * S(x, y). \tag{5}$$

¹ The property itself will henceforth be called the *separability* property.

Since the crisp set is a special case of a fuzzy set, we may easily determine an extensional hull of a crisp set as well. A special case of crisp sets is a singleton that can represent a real number.

Definition 2.5. A fuzzy set $\tilde{x} \in \mathbb{R}$ is called a *singleton determined by* $x \in \mathbb{R}$ if $\tilde{x}(x) = 1$ and $\tilde{x}(y) = 0$ for any $y \in \mathbb{R}$ such that $y \neq x$.

Thus, if we consider a real number $x \in \mathbb{R}$ represented in $\mathcal{F}(\mathbb{R})$ by the singleton $\tilde{x}$, we may construct an extensional hull of x (represented by $\tilde{x}$) with respect to a given similarity relation S . This notion is usually called a *fuzzy point*, and its use has been found to be beneficial in many investigations [18–20].

Indeed, a fuzzy point is in our opinion conceptually the genuine representation of the vaguely defined real number, as it is composed as an extensional hull of the given real number. In other words, it expresses a real number with its neighborhood, i.e., with its class of (fuzzy) equivalence given by the similarity relation S . Thus, we precisely meet the original goal of modeling an expression such as “about five” by dealing with the crisp number 5 and a tolerance of close values (determined by S). Because a fuzzy point could generally be constructed from an arbitrary point in an arbitrary space and not necessarily from a real number, for our purpose, we will again restrict our focus to the universe of discourse $\mathbb{R}$ and rename it an *extensional fuzzy number*.

Definition 2.6. (See [18].) Let S be a $*$ -similarity relation on $\mathbb{R}$, and let $\tilde{x} \in \mathcal{F}(\mathbb{R})$ be a singleton determined by $x \in \mathbb{R}$. The *extensional fuzzy number (fuzzy point)* $x_S \in \mathcal{F}(\mathbb{R})$ is the extensional hull of $\tilde{x}$ w.r.t. S .

For the sake of simplicity and clarity in our notation, we will not distinguish between $x \in \mathbb{R}$ and the singleton $\tilde{x}$ that attains normality at x and will freely talk about an extensional hull of a real number x .

Lemma 2.2. The extensional hull of $x \in \mathbb{R}$ w.r.t. a $*$ -similarity relation S on $\mathbb{R}$ may be expressed as follows:

$$x_S(y) = S(x, y), \quad y \in \mathbb{R}.$$

Proof. By the direct use of (5). $\square$

Though Lemma 2.2 is trivial, it is very important because it shows that dealing with extensional fuzzy numbers is as easy as dealing with chosen similarity relations evaluated in real numbers.

2.2. Elementary but working example

Let us restrict our focus on fuzzy numbers that are extensional w.r.t. a single fixed similarity relation S . We will obtain a very simple but functional example. Consider a separated $*$ -similarity relation S on $\mathbb{R}$ and the set of all fuzzy numbers that are extensional w.r.t. S :

$$\mathcal{F}_S(\mathbb{R}) = \{x_S \mid x \in \mathbb{R}\}.$$

Let us define operations $+$ and $\cdot$ on $\mathcal{F}_S(\mathbb{R})$ as follows:

$$x_S + y_S = (x + y)_S,$$

$$x_S \cdot y_S = (xy)_S,$$

for any $x_S, y_S \in \mathcal{F}_S(\mathbb{R})$. Note that since S is separated, both previous operations are well defined.² Moreover, both operations are obviously commutative. Further, let us define the identity elements for both operations as follows:

$$\mathbf{0} = 0_S = S(0, \cdot) \quad \text{and} \quad \mathbf{1} = 1_S = S(1, \cdot).$$

² Assuming S to be separated, each fuzzy number x_S is uniquely determined by x (see Remark 2.1). If we omitted the assumption of the separability property, we would have to admit $x_S = x'_S$ for two different reals x and x' . In this case, however, $(x + y)_S \neq (x' + y)_S$ in general, and the definition of addition would not be correct. Similarly, the same assumption plays an important role in the correctness of the multiplication operation.

Obviously, the identity elements are well defined, since

$$x_S + \mathbf{0} = x_S + 0_S = (x + 0)_S = x_S,$$

$$x_S \cdot \mathbf{1} = x_S \cdot 1_S = (x \cdot 1)_S = x_S$$

hold for arbitrary $x_S \in \mathcal{F}_S(\mathbb{R})$. Finally, let us define the inverse elements with respect to both operations as follows:

$$-(x_S) = (-x)_S, \quad \forall x_S \in \mathcal{F}_S(\mathbb{R}),$$

$$(x_S)^{-1} = (1/x)_S, \quad \forall x_S \in \mathcal{F}_S(\mathbb{R}) \setminus \{\mathbf{0}\}.$$

Again, one can easily check that

$$x_S + (-x_S) = \mathbf{0}, \quad \forall x_S \in \mathcal{F}_S(\mathbb{R}),$$

$$x_S \cdot (x_S)^{-1} = \mathbf{1}, \quad \forall x_S \in \mathcal{F}_S(\mathbb{R}) \setminus \{\mathbf{0}\}.$$

Please note that the identity elements are not singletons, which is the main difference from the usual arithmetics of fuzzy numbers based on Zadeh's extensional principle, where the calculus directly enforces that only singletons $\tilde{0}$, $\tilde{1}$ may serve as identity elements. However, this insistence is precisely the problem of such arithmetics because it is not possible to determine inverse elements $-a$, a^{-1} that would yield

$$a + (-a) = \tilde{0}, \quad \forall a \in \mathcal{F}(\mathbb{R}),$$

$$a \cdot (a)^{-1} = \tilde{1}, \quad \forall a \in \mathcal{F}(\mathbb{R}) \setminus \{\tilde{0}\}.$$

Let us stress that the fact that $\mathbf{0}$, $\mathbf{1}$ are not singletons is again very natural. If we deal with vaguely defined numbers, identity elements are necessarily elements of the set of vaguely defined numbers, and thus one may hardly expect these identity elements to be precise. Therefore, we may conclude that in the case of arithmetics based on the extensional principle, the fact that calculus enforces the identity elements to be singletons $\tilde{0}$, $\tilde{1}$ is “guilty” of preventing the existence of inverse elements in such a calculus.

Furthermore, one may easily check that in the above motivating example, associativity as well as both distributive laws:

$$(x_S + y_S) \cdot z_S = (x_S \cdot z_S) + (y_S \cdot z_S),$$

$$z_S \cdot (x_S + y_S) = (z_S \cdot x_S) + (z_S \cdot y_S)$$

hold. Therefore, the algebraic structure $(\mathcal{F}_S(\mathbb{R}), +, \cdot, \mathbf{0}, \mathbf{1})$ is a *field*, i.e., the same algebraic structure that is formed by $(\mathbb{R}, +, \cdot, 0, 1)$.

Obviously, there is a one-to-one correspondence between the structures for $\mathcal{F}_S(\mathbb{R})$ and for $\mathbb{R}$, and the example is too simplistic to capture the needs of the intended calculus of fuzzy numbers. However, this structure that naturally grows from fuzzy numbers that are extensional with respect to a single fixed similarity relation serves as a motivating example that shows that:

- the arithmetic based on extensional fuzzy numbers is intuitive, easy to compute and very natural;
- the so far often met problem of the non-existence of inverse elements is only a secondary consequence of the crispness of the identity elements;
- the suggested arithmetical operations based on extensional fuzzy numbers requires the identity elements to be vaguely defined and thus allows for the existence of inverse elements;

which are unquestionably encouraging observations.

Let us finally point out that on the one hand, computations according to the above suggested arithmetic of extensional fuzzy numbers are intuitive because they follow the idea of computing that “about five” plus “about three” is something “about (five plus three)”. This is, in our opinion, much closer to the human style of computing with imprecise numbers than computing with α -cuts or the parameters of functions that express the imprecise numbers. On the other hand, in the suggested arithmetic, the tolerance expressed by “about” is always the same, not only in both summands but also in the computed result, which may seem very unnatural in many cases. Let this issue serve as a motivation for the latter section.

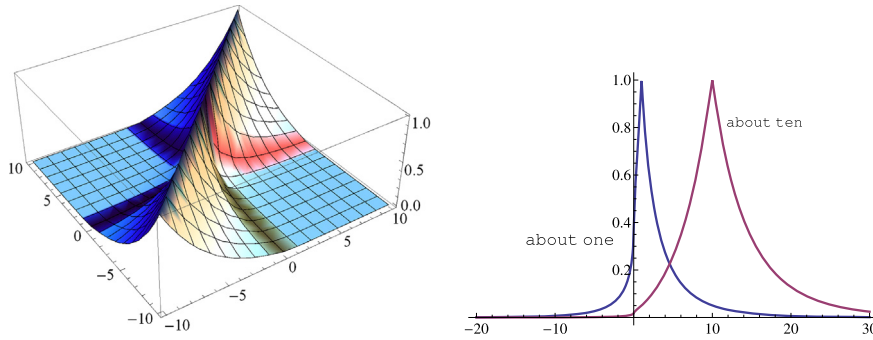


Fig. 1. Non-shift-invariant product type similarity relation S_f ($p = 0.6$) and extensional fuzzy numbers “about one” and “about ten” with respect to S_f .

Remark 2.2. In the above text, we have mentioned that in the simple motivating example, there is a problem of the only measure of tolerance, which might be rather unnatural. However, this is meant in such a way that we deal with the only similarity relation that expresses the tolerance, which does not necessarily mean that we tolerate the same imprecision or vagueness over the whole universe of reals. This objection would be true only in the case of a shift-invariant similarity S , but one may easily construct a similarity S that is not shift-invariant. For example, if we define $f(x) = \text{sgn}(x)|x|^p$, $p > 0$ and $p \neq 1$, then

$$S_f(x, y) = e^{-|f(x) - f(y)|}$$

is a $*$ -similarity relation $\mathbb{R}$, where $*$ is the product t-norm, but is not a shift-invariant similarity. As one can see from Fig. 1, such similarity allows us to consider the scale of compared values, e.g., $S(1, 2) = 0.597$, but $S(10, 11) = 0.791$, which is very natural. Thus, even the simple example with a single similarity captures a type of desirable flexibility.

3. Arithmetics of extensional fuzzy numbers

3.1. Conceptual approach – additional notions

The drawback of the above introduced example of the arithmetic of fuzzy numbers that are extensional w.r.t. a single similarity relation is obvious – the only tolerance or vagueness measure. However, there is a natural way out of it: employing a whole system of similarity relations in the suggested arithmetic. For this purpose, we need to define the transitive closure of similarity relations (see [3,8,10]).

For the rest of the paper, we set the following notation: for two arbitrary binary fuzzy relations S, S' on a universe $X \times Y$, $S \subseteq S'$ denotes the fact that

$$S(x, y) \leq S'(x, y) \quad \text{for any } (x, y) \in X \times Y.$$

Definition 3.1. Let $*$ be a left-continuous t-norm, and let S be a reflexive and symmetric binary fuzzy relation on $\mathbb{R}$. Then, the binary fuzzy relation on $\mathbb{R}$ denoted by $\text{cl}(S)$ is the *transitive closure* of S if it satisfies:

- (i) $\text{cl}(S)$ is a $*$ -similarity relation,
- (ii) $S \subseteq \text{cl}(S)$,
- (iii) for any $*$ -similarity relation S' on $\mathbb{R}$, if $S \subseteq S'$ then $\text{cl}(S) \subseteq S'$.

Since the intersection of $*$ -similarity relations is again a $*$ -similarity relation, one can check that

$$\text{cl}(S) = \bigcap \{T \mid T \text{ is } * \text{-similarity relation and } S \subseteq T\}, \quad (6)$$

where

$$\left(\bigcap_{i \in I} S_i \right)(x, y) = \bigwedge_{i \in I} S_i(x, y)$$

for any $x, y \in \mathbb{R}$. Obviously, if S is a $*$ -similarity relation then $\text{cl}(S) = S$. In the other case, the transitive closure may be determined as follows.

Lemma 3.1. (See [8].) *Let $*$ be a left-continuous t -norm, and let S be a reflexive and symmetric binary fuzzy relation on $\mathbb{R}$. Then, for any $x, y \in \mathbb{R}$, the following holds:*

$$\text{cl}(S)(x, y) = \left(\bigcup_{n=1}^{\infty} S^{(n)} \right)(x, y) = \bigvee_{n=1}^{\infty} S^{(n)}(x, y),$$

where

$$\begin{aligned} S^{(1)}(x, y) &= S(x, y), \\ S^{(n)}(x, y) &= \bigvee_{z \in \mathbb{R}} (S^{(1)}(x, z) * S^{(n-1)}(z, y)), \quad n \geq 2. \end{aligned}$$

Finally, we may consider the transitive closure of the union of two given similarity relations to define a union in the set of extensional $*$ -similarity relations:

$$S \cup T = \text{cl}(S \cup T).$$

The following lemma summarizes the important properties of $\cup$ used to verify the properties of arithmetical operations on extensional fuzzy numbers.

Lemma 3.2. *The operation $\cup$ is idempotent, commutative, and associative and preserves the shift-invariance of $*$ -similarity relations.*

Proof. Obviously, $\cup$ is idempotent and commutative. Let $\text{Cls}(S)$ denote the set of all supersets of S that are $*$ -similarity relations. By Definition 3.1, we have $\text{cl}(S) = \bigcap \text{Cls}(S)$. Let R, S, T be $*$ -similarity relations. To prove the associativity of $\cup$, it suffices to check the following equality:

$$\text{Cls}(R \cup (S \cup T)) = \text{Cls}((R \cup S) \cup T). \quad (7)$$

If $H \in \text{Cls}(R \cup (S \cup T))$, then $R \cup (S \cup T) \subseteq H$, which implies $R \cup S \cup T \subseteq H$. Since $R \cup S \subseteq H$, we obtain $R \cup S \subseteq H$, whence $(R \cup S) \cup T \subseteq H$. Thus, $H \in \text{Cls}((R \cup S) \cup T)$ and $\text{Cls}(R \cup (S \cup T)) \subseteq \text{Cls}((R \cup S) \cup T)$. Analogously, one may check the opposite inclusion, and (7) is proved.

Let S, T be shift-invariant $*$ -similarity relations, and let $H = S \cup T$. Obviously, H is reflexive, symmetric, and shift-invariant. We will check that the construction of the transitive closure introduced in Lemma 3.1 preserves the shift-invariance of the fuzzy relations. By induction, $H^{(1)} = H$ is shift-invariant. Suppose that the same holds for $H^{(n)}$. We will prove that $H^{(n+1)}$ is shift-invariant. Let $x, y, h \in \mathbb{R}$. Then,

$$\begin{aligned} H^{(n+1)}(x, y) &= \bigvee_{t \in \mathbb{R}} (H^{(1)}(x, t) * H^{(n)}(t, y)) \\ &= \bigvee_{t \in \mathbb{R}} (H^{(1)}(x + h, t + h) * H^{(n)}(t + h, y + h)) \\ &= \bigvee_{s \in \mathbb{R}} (H^{(1)}(x + h, s) * H^{(n)}(s, y + h)) = H^{(n+1)}(x + h, y + h). \end{aligned}$$

Thus, $H^{(n)}$ is shift-invariant for any $n = 1, 2, \dots$. Hence,

$$\text{cl}(H)(x + h, y + h) = \bigvee_{i=1}^{\infty} H^{(i)}(x + h, y + h) = \bigvee_{i=1}^{\infty} H^{(i)}(x, y) = \text{cl}(H)(x, y),$$

which concludes the proof. $\square$

Lemma 3.3. Let $\{S_i \mid i \in I\}$ be a non-empty set of shift-invariant $*$ -similarity relations on $\mathbb{R}$. Then,

$$S(x, y) = \bigwedge_{i \in I} S_i(x, y) \quad (8)$$

is a shift-invariant $*$ -similarity relation.

Proof. Let $\{S_i \mid i \in I\}$ be a non-empty set of shift-invariant $*$ -similarity relations on $\mathbb{R}$ and $x, y, h \in \mathbb{R}$. Since $S_i(x, y) = S_i(x + h, y + h)$, we obtain

$$\begin{aligned} S(x + h, y + h) &= \left(\bigcap_{i \in I} S_i \right)(x + h, y + h) = \bigwedge_{i \in I} S_i(x + h, y + h) \\ &= \bigwedge_{i \in I} S_i(x, y) = \left(\bigcap_{i \in I} S_i \right)(x, y) = S(x, y), \end{aligned}$$

which concludes the proof. $\square$

3.2. Closed system of $*$ -similarity relations – advanced example

The next natural step is to extend the motivating example of a functional arithmetic of fuzzy numbers that are extensional w.r.t. a similarity relation to arithmetics of fuzzy numbers that are extensional w.r.t. a similarity from a given system of similarity relations $\mathcal{S}$. Such an extension would allow us to deal with distinct “measures of vagueness or tolerance” from the given system.

Definition 3.2. Let $\mathcal{S}$ be a non-empty set of $*$ -similarity relations and denote

$$S^* = \bigcap_{S \in \mathcal{S}} S. \quad (9)$$

The set $\mathcal{S}$ is said to be a *closed system of $*$ -similarity relations* provided that

- (i) if $S, T \in \mathcal{S}$ and $S(x, \cdot) = T(y, \cdot)$, then $x = y$ and $S = T$,
- (ii) $S^* \in \mathcal{S}$,
- (iii) $S \cup T \in \mathcal{S}$ for arbitrary $S, T \in \mathcal{S}$.

Note that the condition (i) in the previous definition generalizes the separability property for $*$ -similarity relations to an analogous property for systems of $*$ -similarity relations. In the example of Section 2.2, the separability property was used to ensure the correctness of arithmetic operations for fuzzy numbers related to one $*$ -similarity relation, and, as will be seen below, condition (i) plays the same role. The following lemma shows that (i) is satisfied, assuming systems of shift-invariant separated $*$ -similarity relations.

Lemma 3.4. Let $\mathcal{S}$ be a non-empty set of shift-invariant separated $*$ -similarity relations. Then, $\mathcal{S}$ is a closed system of $*$ -similarity relations if and only if (ii) and (iii) of Definition 3.2 are satisfied.

Proof. $(\Rightarrow)$ Trivial. $(\Leftarrow)$ Let $\mathcal{S}$ be a non-empty set of shift-invariant, separated $*$ -similarity relations that satisfies (ii) and (iii) of Definition 3.2. Let $S, T \in \mathcal{S}$ and $x, x' \in \mathbb{R}$ be such that $S(x, \cdot) = T(x', \cdot)$. First, we will show that $x = x'$. Since S, T are separated, and $S(x, y) = T(x', y)$ holds for any $y \in \mathbb{R}$, then, setting $y = x$, we obtain $T(x', x) = 1$, which implies $x = x'$. Let $y, z \in \mathbb{R}$ be arbitrary and set $h = y - x$. Since S, T are shift-invariant and $S(x, z - h) = T(x', z - h) = T(x, z - h)$, then $S(y, z) = S(x + h, z - h + h) = T(x + h, z - h + h) = T(y, z)$, which concludes the proof. $\square$

In the sequel, let $\mathcal{S}$ be a closed system of $*$ -similarity relations, and let $\mathcal{F}_{\mathcal{S}}(\mathbb{R})$ denote the set of all fuzzy numbers that are extensional w.r.t. $*$ -similarities from $\mathcal{S}$, i.e.,

$$\mathcal{F}_{\mathcal{S}}(\mathbb{R}) = \{x_S \mid x \in \mathbb{R} \text{ and } S \in \mathcal{S}\}.$$

The following theorem shows one way through which the summation and multiplication of reals can be extended to extensional fuzzy numbers from $\mathcal{F}_S(\mathbb{R})$.

Theorem 3.5. *The triplets $(\mathcal{F}_S(\mathbb{R}), +, \mathbf{0})$ and $(\mathcal{F}_S(\mathbb{R}), \cdot, \mathbf{1})$, where the operations $+$ and $\cdot$ on $\mathcal{F}_S(\mathbb{R})$ are defined by*

$$x_S + y_T = (x + y)_{S \cup T}, \quad x_S \cdot y_T = (x \cdot y)_{S \cup T}, \quad S, T \in \mathcal{S},$$

and the identity elements $\mathbf{0}$ and $\mathbf{1}$ by

$$\mathbf{0} = 0_{S^*} = S^*(0, \cdot), \quad \mathbf{1} = 1_{S^*} = S^*(1, \cdot),$$

form abelian monoids. Moreover, the following distributive law:

$$(x_R + y_S) \cdot z_T = (x_R \cdot z_T) + (y_S \cdot z_T)$$

holds for any $R, S, T \in \mathcal{S}$.

Proof. By the condition (i) of [Definition 3.2](#), each fuzzy number x_S is uniquely determined by x and S , i.e., $x_S = x'_T$ if and only if $x = x'$ and $S = T$. Therefore, both operations are well defined. Using [Lemma 3.2](#), both operations are commutative and associative. Furthermore, we have

$$x_S + \mathbf{0} = x_S + 0_{S^*} = (x + 0)_{S \cup S^*} = x_S,$$

$$x_S \cdot \mathbf{1} = x_S \cdot 1_{S^*} = (x \cdot 1)_{S \cup S^*} = x_S,$$

which concludes the first part of the theorem. The second part can be proved directly as follows:

$$\begin{aligned} (x_R + y_S) \cdot z_T &= (x + y)_{R \cup S} \cdot z_T \\ &= ((x + y) \cdot z)_{(R \cup S) \cup T} = (x \cdot z + y \cdot z)_{(R \cup T) \cup (S \cup T)} \\ &= (x \cdot z)_{R \cup T} + (y \cdot z)_{S \cup T} = (z_T \cdot x_R) + (z_T \cdot y_S), \end{aligned}$$

where we used the distributive law for real numbers and the idempotency and associativity of $\cup$, which hold due to [Lemma 3.2](#). $\square$

Now, if we approach the inverse elements like we do the elementary motivating example, i.e., if

$$-(x_S) = (-x)_S, \quad \forall x_S \in \mathcal{F}_S(\mathbb{R}),$$

$$(x_S)^{-1} = (1/x)_S, \quad \forall x_S \in \mathcal{F}_S(\mathbb{R}) \setminus \{0_S \mid S \in \mathcal{S}\},$$

we will encounter the following problems:

$$x_S + (-x_S) = 0_S \neq 0_{S^*} = \mathbf{0},$$

$$x_S \cdot (x_S)^{-1} = 1_S \neq 1_{S^*} = \mathbf{1}$$

for all $S \in \mathcal{S}$ such that $S \neq S^*$. Thus, the inverse elements do not work as well as the classical inverse element and do not give us the identity elements $\mathbf{0}$ and $\mathbf{1}$, respectively. However, at least the obtained elements $0_S, 1_S$ are constructed with help of $x = 0$ and $x = 1$, respectively, and thus they differ from the identity elements $\mathbf{0}$ and $\mathbf{1}$ only in the measure of vagueness given by the similarity relation S .

In other words, $(\mathcal{F}_S(\mathbb{R}), +, \cdot, \mathbf{0}, \mathbf{1})$ is not a field but is an algebraic structure that is very close to a field. The question of what kind of algebraic structure it is answered in the next section. One can see that considering a set of identity elements instead of a single identity for each operations is the most natural way out of this problem.

Remark 3.1. It should be noted that the idea of dealing with a set of identity elements is not completely new. In [\[27\]](#), M. Mareš has recognized the importance of symmetric fuzzy quantities; therefore, he introduced sets of additive and multiplicative symmetric elements and interesting equivalences stating that two fuzzy quantities are equal up

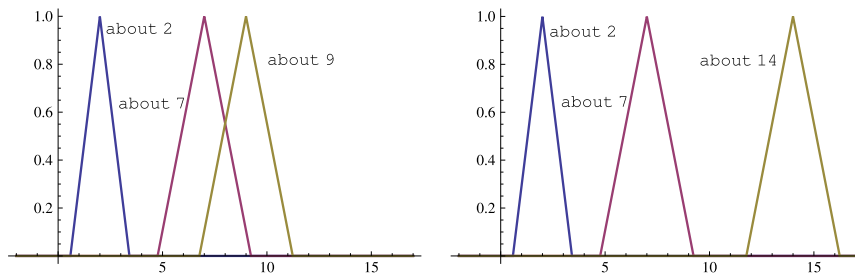


Fig. 2. Summation and multiplication of the extensional fuzzy numbers “about 2” ($S_{0.5}$) and “about 7” ($S_{0.1}$).

to symmetric fuzzy quantities.³ S. Markov [29,30] has considered a set of all so-called symmetric convex bodies (intervals) in his development of q -linear spaces, especially in characterizing the solutions of algebraic equations. A. Bica [4] has pointed out sets containing zero-symmetric and one-symmetric elements (L-R fuzzy sets) that form special submonoids of commutative monoids of fuzzy numbers.⁴

Example 3.1. Let $*$ be the Łukasiewicz t-norm [21,37] and consider the following closed system of embedded shift-invariant separated $*$ -similarities:

$$\mathcal{S} = \{S_p \mid p \in [a, b] \text{ and } S_p(x, y) = (1 - p|x - y|) \vee 0\},$$

where $0 < a \leq b$. Since $S_p \cup S_{p'} = S_p$ for $p \leq p'$, and since all similarities of $\mathcal{S}$ are transitive, it also holds that

$$S_p \cup S_{p'} = S_p, \quad \text{for } p \leq p'.$$

Therefore,

$$x_{S_p} + y_{S_{p'}} = (x + y)_{S_{p''}}, \quad x_{S_p} \cdot y_{S_{p'}} = (x \cdot y)_{S_{p''}}, \quad p'' = \min\{p, p'\}.$$

Finally, $S^* = \bigcap \mathcal{S} = S_b$, and therefore,

$$\mathbf{0} = 0_{S_b} = S_b(0, \cdot), \quad \mathbf{1} = 1_{S_b} = S_b(1, \cdot),$$

where $S_b(x, y) = (1 - b|x - y|) \vee 0$. In Fig. 2, the summation and multiplication of “about 2” and “about 7” are depicted. One can see that the resulting fuzzy sets “about 9” and “about 14” retain the fuzziness of “about 7” because “about 7” contains the fuzziness of “about 2” (i.e., $S_{0.1} \subseteq S_{0.5}$).

Example 3.1 describes the arithmetic of triangular shaped fuzzy numbers that are very common. It should be stressed that in comparison to the often used arithmetics of such fuzzy numbers that stem from Zadeh’s extensional principle or that are α -cut based, there are at least three unquestionable advantages:

- the triangular shape is preserved even in the case of the multiplication or the division, which is not only intuitive but also highly desirable (the fuzzy number is still a vaguely defined number with some tolerance of the vagueness);
- the computational efforts are very low, as computing the central point is computationally as cheap as in the case of real numbers, and the only possibly expensive step – the computation of the transitive closure of two involved similarities – is in this case as cheap as comparing p and p' ;
- the obtained triangular fuzzy numbers do not grow indefinitely wide.

We briefly discuss all three items. The first presents an unquestionable advantage. Of course, our example does not provide the only approach that can ensure the preservation of a given shape of fuzzy numbers. However, most of the

³ Further remarks on M. Mareš’s work may be found in Section 7.

⁴ Practically, A. Bica introduced an algebraic conception of an arithmetic of fuzzy numbers that is similar to our approach presented in the following section. Nevertheless, it should be stressed that Bica’s ideas are more restrictive, and therefore, they can be simply expressed by the tools presented below.

known approaches that may serve for this task are designed rather as technical modifications of the existing arithmetic that does not preserve a given shape, than as theoretical constructions.

One may argue that the second argument in favor of the suggested approach is not that significant, as the standard approach based on Zadeh's extensional principle is very efficient if it is not applied in the α -cut based scheme but with the use of parameters for the triangular fuzzy numbers. This is true, but only in the case of summation and for this particular shape of fuzzy sets. Let us consider a rather simple but yet computationally expensive example in the case of Zadeh's standard extensional principle-based approach (especially for the multiplication). This example of arithmetic is still as computationally cheap as the one given in [Example 3.1](#), if we use the suggested approach based on extensional fuzzy numbers.

The last argument may seem to some readers most counterintuitive, as an extended literature on the propagation of uncertainty has been published, and it usually assumes an increase in fuzziness by the use of arithmetic operations. This view must be respected, and we do not question it in general. However, many real-world problems may lead to other perspectives, i.e., to calculating with objects with significantly different fuzziness or imprecisions. In such a case, the greater imprecision contained in one of the summands (multiplicands) already contains the smaller imprecision contained in the other summand (multiplicand) that no longer affects the final imprecision anymore. A similar idea is mathematically justified, e.g., in counting with quantities with so-called heavy tailed distributions of measurement. For more details, we refer readers to [\[25\]](#).

Example 3.2. Let $*$ be the product t-norm [\[21,37\]](#), and consider the following closed system of embedded shift-invariant separated $*$ -similarities:

$$\mathcal{S} = \{S_p \mid p \in [a, b] \text{ and } S_p(x, y) = e^{-p|x-y|}\},$$

where $0 < a \leq b$. Again, since $S_p \cup S_{p'} = S_p$ for $p \leq p'$, and since all similarities of $\mathcal{S}$ are transitive, it also holds that

$$S_p \cup S_{p'} = S_p, \quad \text{for } p \leq p',$$

and thus as in the previous example:

$$x_{S_p} + y_{S_{p'}} = (x + y)_{S_{p''}}, \quad x_{S_p} \cdot y_{S_{p'}} = (x \cdot y)_{S_{p''}}, \quad p'' = \min\{p, p'\}.$$

Finally, $\mathcal{S}^* = \bigcap \mathcal{S} = S_b$, and therefore,

$$\mathbf{0} = 0_{S_b} = S_b(0, \cdot), \quad \mathbf{1} = 1_{S_b} = S_b(1, \cdot),$$

where $S_b(x, y) = e^{-b|x-y|}$.

[Example 3.2](#) describes the arithmetic that deals with the Gaussian-shaped⁵ fuzzy sets. Again, as in the case of the arithmetic described in [Example 3.1](#), the computation for multiplication (and hence division too) is even computationally as cheap as in the case of real numbers, and the only possibly expensive step – the computation of the transitive closure of two involved similarities – is in this case as cheap as the comparison of p and p' . Again, it can be stated that the resulting fuzzy sets do not grow indefinitely wide.

Remark 3.2. Note that because all $S_p \in \mathcal{S}$ from [Examples 3.1 and 3.2](#) are shift-invariant, it holds that $S_p(1, 2) = S_p(10001, 10002)$. However, the flexibility is not lost at all, but is provided by the variety of similarity relations available in the system $\mathcal{S}$, and thus one can easily model a similarity of small values by an S_p for some high p and a similarity of high values by an $S_{p'}$ for some small p' .

Working with extensional fuzzy numbers, we need not restrict ourselves to the arithmetic determined by the union of $*$ -similarity relations. The following example demonstrates an alternative based on addition and multiplication with real numbers.

⁵ We are fully aware of the fact that (due to their unlimited support) Gaussian-shaped fuzzy sets do not fit into most of the schemes of fuzzy numbers. However, we do not see any theoretical or conceptual reason for that exclusion, and we are convinced that this restriction is only of a technical nature. On the contrary, in our opinion such fuzzy sets perfectly fit into a scheme for computing with vaguely defined numbers.

Example 3.3. Let $*$ be the Łukasiewicz t-norm and

$$\mathcal{R} = \{R_p \mid p \in [0, \infty) \text{ and } R_p(x, y) = (1 - p|x - y|) \vee 0\}$$

be the closed system of embedded shift-invariant $*$ -similarity relations (cf., Example 3.1). Define

$$\begin{aligned} x_{R_p} \boxplus y_{R_{p'}} &= (x + y)_{R_{p+p'}}, & \mathbf{0} &= 0_{R_0} = R_0(0, \cdot), \\ x_{R_p} \boxdot y_{R_{p'}} &= (x \cdot y)_{R_{p \cdot p'}}, & \mathbf{1} &= 1_{R_1} = R_1(1, \cdot). \end{aligned}$$

It is easy to check that $(\mathcal{FR}(\mathbb{R}), \boxplus, \mathbf{0})$ and $(\mathcal{FR}(\mathbb{R}), \boxdot, \mathbf{1})$ form abelian monoids and, moreover, that the distributivity law is satisfied. Nevertheless, it should be noted that this definition does not propagate the vagueness presented in extensional fuzzy numbers well because a resulting fuzzy number may express less vagueness than was present in the related addends or multiplicands.

For other details related to the proposed arithmetics of extensional fuzzy numbers, we refer to [13].

4. MI-monoids

4.1. Main definition and examples

In the above sections, we have introduced a novel approach to arithmetics of extensional fuzzy numbers that seems to work well from an algebraic point of view, i.e., to be close to the field structure. However, one substantial drawback seems to be present, which is that for $S \neq S^*$, the following holds:

$$x_S \cdot (x_S)^{-1} = 1_S \neq \mathbf{1}.$$

Let us focus on the properties of such 1_S . We may see that

$$\begin{aligned} 1_S &= (1_S)^{-1}, \\ 1_S \cdot x_S &= x_S \cdot 1_S, \end{aligned}$$

which are exactly the properties of $\mathbf{1} = 1_{S^*}$. In other words, 1_S is an identity-like element. The same can be shown about 0_S in the case of summation.

Therefore, we naturally introduce an algebraic structure that contains a set of identity elements. The introduced structure will be called an MI-monoid, where MI stands for *Many Identities*.

Definition 4.1. A triplet $(G, \circ, E)$ is said to be an *MI-monoid* if E is a non-empty subset of G and $\circ$ is a binary operation on G such that, for any $x, y, z \in G$ and $a, b \in E$, the following conditions are satisfied:

- (M1) $x \circ (y \circ z) = (x \circ y) \circ z$,
- (M2) $\exists e \in E, \forall x \in G : x \circ e = e \circ x = x$,
- (M3) $a \circ b \in E$,
- (M4) $x \circ a = a \circ x$.

Elements from E are called *pseudoidentity elements* (*pseudoidentities*, for short), and a pseudoidentity e satisfying (M2) is called a (*strong*) *identity element*. An MI-monoid G is said to be *abelian* or *commutative* if $x \circ y = y \circ x$ holds for any $x, y \in G$.

Standardly, we write $G = (G, \circ, E)$ and $x \circ y = xy$. Clearly, for the abelian MI-monoids, the axiom (M4) may be omitted. One can easily verify that each MI-monoid possesses a unique identity element $e \in E$, i.e., a single pseudoidentity satisfies (M2). In what follows, the identity element will be always denoted by e . Note that, in contrast to exactly one identity element, an MI-monoid can have an arbitrary number of pseudoidentity elements (including no pseudoidentity element except the strong identity, i.e., $E \setminus \{e\} = \emptyset$). A larger number of pseudoidentities in G could be interpreted as more “fuzziness” in the elements from G .

Definition 4.2. An MI-monoid G is said to be *good* if the following condition is satisfied:

$$(M5) \quad \forall x \in G, \forall a \in E: \text{ if } x \circ a \in E \text{ then } x \in E.$$

If G is a good MI-monoid, then we need not suppose $e \in E$ in (M2), but it suffices to consider $e \in G$. In fact, by (M2), we have $ae = a$ for any $a \in E$. Hence, $e \in E$ by (M5).

Remark 4.1. Obviously, one can simply check that $(G, \circ, E)$ is an MI-monoid if and only if $(G, \circ)$ is a monoid, $(E, \circ)$ is an abelian submonoid of G , and arbitrary $x \in G$ and $a \in E$ commute. It is easy to see that each monoid G is an MI-monoid with $E = \{e\}$.

One can check that $(\mathcal{F}_S(\mathbb{R}), +, E_S^0)$, where $E_S^0 = \{0_S \mid S \in \mathcal{S}\}$, is an MI-monoid with the strong identity element 0_{S^*} , and similarly, $(\mathcal{F}_S(\mathbb{R}), \cdot, E_S^1)$, where $E_S^1 = \{1_S \mid S \in \mathcal{S}\}$, is an MI-monoid with the strong identity element 1_{S^*} . Similarly, one could define MI-monoids replacing $\mathcal{S}$ by the closed system $\mathcal{R}$ introduced in Example 3.3.

Moreover, it can be shown that all examples of MI-monoids of extensional fuzzy numbers determined by closed systems $\mathcal{S}$ and $\mathcal{R}$ are also good MI-monoids. However, is there a non-trivial example of an MI-monoid that is not good? The answer is positive, and we provide readers with two very natural examples of such MI-monoids.

Example 4.1. Let $G = \{[a, b] \mid a, b \in \mathbb{R}, a \leq b\}$ be the set of all real intervals, $E^0 = \{[a, b] \mid a, b \in \mathbb{R}, a \leq b \text{ and } 0 \in [a, b]\}$, and let the operation $+$ on G be defined as follows:

$$[a, b] + [c, d] = [a + c, b + d].$$

Then, the strong identity element is $[0, 0]$, and $(G, +, E^0)$ is an additive abelian MI-monoid of real intervals that is not good because, e.g.,

$$[-5, 5] + [1, 2] = [-4, 7],$$

but $[1, 2] \notin E^0$. Similarly, let G be defined as above, E^1 denote the set of all intervals containing 1 and the operation $\cdot$ be defined as follows

$$[a, b] \cdot [c, d] = [\min(ac, ad, bc, bd), \max(ac, ad, bc, bd)].$$

Then, the strong identity element is $[1, 1]$, and $(G, \cdot, E^1)$ is an abelian multiplicative MI-monoid that is not good because, e.g.,

$$[0.2, 2] \cdot [2, 4] = [0.4, 8],$$

but $[2, 4] \notin E^1$.

Of course, both examples of non-good MI-monoids can be enhanced by a redefinition of the sets of pseudoidentity elements in such a way that they contain only intervals symmetrically distributed around 0 or 1.

The suitability of the MI-monoid structure may also be demonstrated by the fact that it includes an algebraic framework for the standard approach to the arithmetic of fuzzy numbers (cf. [9,24]). This fact is demonstrated in the following example.

Example 4.2. Let G be the MI-monoid of real intervals (see Example 4.1), and let H be the set of all mappings $g : [0, 1] \rightarrow G$ provided

$$\begin{aligned} g(1) &\neq \emptyset, \\ g(\alpha) &\subseteq g(\beta) \quad \text{for all } \alpha, \beta \in [0, 1], \alpha \geq \beta, \\ \bigcup_{\substack{\beta \in [0, 1] \\ \alpha < \beta}} g(\beta) &= g(\alpha) \quad \text{for all } \alpha \in [0, 1]. \end{aligned} \tag{10}$$

One can see that g represents an upper semicontinuous and convex fuzzy set in $\mathbb{R}$ with a bounded closed support $g(0)$ (a fuzzy number). More precisely, if $g \in H$, then

$$A_g(x) = \bigvee_{\substack{\alpha \in [0,1], \\ x \in g(\alpha)}} \alpha$$

is an upper semicontinuous and convex fuzzy set $A_g : \mathbb{R} \rightarrow [0, 1]$. Let E^0 be the set of all g such that $0 \in g(0)$, and define pointwise $g + g'$ using the addition of real intervals. Then, $(H, +, E^0)$ is an abelian additive MI-monoid of fuzzy numbers. Analogously, let E^1 be the set of all g such that $1 \in g(0)$, and define pointwise $g \cdot g'$ using the multiplication of real intervals. Then, $(H, \cdot, E^1)$ is an abelian multiplicative MI-monoid of fuzzy numbers.

In practice, fuzzy numbers are often represented by several parameters to simplify their notation and mainly their arithmetic [24,40]. It should be noted that the simplified arithmetic is only an approximation of the standard arithmetic of fuzzy numbers mentioned in the previous example.

Example 4.3. Let T be the set of all trapezoidal fuzzy numbers, i.e., fuzzy numbers given in a parametric form $A = \langle a_A, b_A, c_A, d_A \rangle$, where $a_A, b_A, c_A, d_A \in \mathbb{R}$ with $a_A \leq b_A \leq c_A \leq d_A$, and defined as

$$A(x) = \begin{cases} 0, & x < a_A \text{ or } d_A < x; \\ (x - a_A)/(b_A - a_A), & a_A \leq x < b_A; \\ 1, & b_A \leq x \leq c_A; \\ (d_A - x)/(d_A - c_A), & c_A < x \leq d_A. \end{cases}$$

Note that although we use the term “trapezoidal”, the set of these fuzzy numbers also contains triangular fuzzy numbers (i.e., $b_A = c_A$) and real intervals (i.e., $a_A = b_A$ and $c_A = d_A$) including one-element intervals (singletons representing the real numbers). Define the addition on T by

$$\begin{aligned} A + B &= \langle a_A, b_A, c_A, d_A \rangle + \langle a_B, b_B, c_B, d_B \rangle \\ &= \langle a_A + a_B, b_A + b_B, c_A + c_B, d_A + d_B \rangle. \end{aligned}$$

Let E^0 be the set of all trapezoidal fuzzy numbers A for which $0 \in (a_A, d_A)$, i.e., A is a pseudoidentity element iff $A(0) > 0$. Clearly, $\langle 0, 0, 0, 0 \rangle$ is the identity element. One may simply check that $(T, +, E^0)$ is a non-good abelian additive MI-monoid. Further, define the multiplication on T by

$$A \cdot B = \langle a_A, b_A, c_A, d_A \rangle \cdot \langle a_B, b_B, c_B, d_B \rangle = \langle a_{A \cdot B}, b_{A \cdot B}, c_{A \cdot B}, d_{A \cdot B} \rangle,$$

where

$$\begin{aligned} a_{A \cdot B} &= \min(a_A a_B, a_A d_B, d_A a_B, d_A d_B), \\ b_{A \cdot B} &= \min(b_A b_B, b_A c_B, c_A b_B, c_A c_B), \\ c_{A \cdot B} &= \max(b_A b_B, b_A c_B, c_A b_B, c_A c_B), \\ d_{A \cdot B} &= \max(a_A a_B, a_A d_B, d_A a_B, d_A d_B). \end{aligned}$$

Let E^1 be the set of all trapezoidal fuzzy numbers for which $1 \in (a_A, d_A)$, i.e., A is a pseudoidentity element iff $A(1) > 0$. Clearly, $\langle 1, 1, 1, 1 \rangle$ is the identity element. Again, one may verify that $(T, \cdot, E^1)$ is a non-good abelian multiplicative MI-monoid.

A recent stream in the development of fuzzy number theory is closely related to the parametric representation of fuzzy numbers using monotonic interpolation introduced by Guerra and Stefanini in [12]. The advantage of this type of parametric representation lies in the additional parameters to characterize more complex shapes of fuzzy numbers. These more complex functions provides an improvement of the approximation of a fuzzy number from data, and importantly, a simple arithmetic based on manipulations of the parameters can be introduced for them.

Example 4.4. In [12] (see also [40]), a parametric representation of fuzzy numbers using monotonic interpolation is provided, and, for instance, the trapezoidal fuzzy numbers constitute an example of such a parametric representation. Let

$$0 = \alpha_0 < \alpha_1 < \dots < \alpha_N = 1 \quad (11)$$

be real numbers for a finite decomposition of the unit interval. Without detailed comment, considering the differentiable case, an LU-fuzzy number A is represented by the following system of vectors

$$A = (\alpha_i; u_i^-, \delta u_i^-, u_i^+, \delta u_i^+)_{i=0, \dots, N}, \quad (12)$$

with the data

$$u_0^- \leq u_1^- \leq \dots \leq u_N^- \leq u_N^+ \leq u_{N-1}^+ \leq \dots \leq u_0^+$$

and the slopes

$$\delta u_i^- \geq 0 \quad \text{and} \quad \delta u_i^+ \leq 0, \quad \text{for all } i = 0, \dots, N.$$

Denote by L the set of all LU-fuzzy numbers in the form (12) for a fixed finite decomposition of $[0, 1]$ determined by (11). Define the addition on L by (α_i is omitted here for simplicity)

$$\begin{aligned} & (u_i^-, \delta u_i^-, u_i^+, \delta u_i^+)_{i=0, \dots, N} + (v_i^-, \delta v_i^-, v_i^+, \delta v_i^+)_{i=0, \dots, N} \\ &= (u_i^- + v_i^-, \delta u_i^- + \delta v_i^-, u_i^+ + v_i^+, \delta u_i^+ + \delta v_i^+)_{i=0, \dots, N}. \end{aligned}$$

Let E^0 be the set of all LU-fuzzy numbers of L such that $0 \in [u_0^-, u_0^+]$. By the results in [40], $(L, +, E^0)$ is a non-good abelian additive MI-monoid. Further, define the multiplication on L by

$$(u_i^-, \delta u_i^-, u_i^+, \delta u_i^+)_{i=0, \dots, N} \cdot (v_i^-, \delta v_i^-, v_i^+, \delta v_i^+)_{i=0, \dots, N} = ((uv)_i^-, \delta (uv)_i^-, (uv)_i^+, \delta (uv)_i^+)_{i=0, \dots, N},$$

where

$$\begin{aligned} (uv)_i^- &= \min(u_i^- v_i^-, u_i^- v_i^+, u_i^+ v_i^-, u_i^+ v_i^+), \\ (uv)_i^+ &= \max(u_i^- v_i^-, u_i^- v_i^+, u_i^+ v_i^-, u_i^+ v_i^+) \end{aligned}$$

and

$$\begin{aligned} (uv)_i^- &= \delta u_i^{p_i^-} v_i^{q_i^-} + u_i^{p_i^-} \delta v_i^{q_i^-} \\ (uv)_i^+ &= \delta u_i^{p_i^+} v_i^{q_i^+} + u_i^{p_i^+} \delta v_i^{q_i^+} \end{aligned}$$

with (p_i^-, q_i^-) being the pair associated with the combination of superscripts $+$ and $-$ giving the minimum $(uv)_i^-$, and (p_i^+, q_i^+) being the pair associated with the combination of $+$ and $-$ giving the maximum $(uv)_i^+$. Let E^1 be the set of all LU-fuzzy numbers from L such that $1 \in [u_0^-, u_0^+]$. By the results in [40], $(L, \cdot, E^1)$ is a non-good abelian multiplicative MI-monoid.

Although Examples 4.2, 4.3 and 4.4 present MI-monoids of fuzzy numbers that are not good, one may easily redefine E^0 and E^1 in such a way that the modified MI-monoids will gain this property. For instance, redefining E^0 in Example 4.3 as

$$E^0 = \{ \langle a, 0, 0, d \rangle \mid a, d \in \mathbb{R}, a \leq 0 \leq d \},$$

we obtain a good additive MI-monoid of trapezoidal fuzzy numbers.⁶ In the definition of a field, the multiplicative group is defined on the support of the abelian additive group, where the zero element is omitted. This idea is also retained in our definition of MI-field (see Definition 6.1) and other MI algebraic structures. Therefore, in some cases, it is appropriate to consider structures that are not good in a sense of Definition 4.2. At the end of this section, we would like to show two examples of MI-monoids that are not purely fuzzy and generalize the interval arithmetic discussed in Example 4.1.

⁶ Indeed, if $A = \langle a, 0, 0, d \rangle$ and $B = \langle a', b', c', d' \rangle$ such that $A + B \in E^0$, then $0 + b' = 0$ and $0 + c' = 0$, which implies $b' = c' = 0$, and $B \in E^0$.

Example 4.5. Let M be the set of all convex compact subsets (or also convex bodies) of $\mathbb{R}^n$, and let $\mathbf{a} + \mathbf{b}$ denote the addition of two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$. Let $\oplus$ be the Minkowski addition, i.e.,

$$A \oplus B = \{\mathbf{a} + \mathbf{b} \mid \mathbf{a} \in A, \mathbf{b} \in B\}.$$

It is easy to see that $\oplus$ is commutative and associative and that the set containing only the null vector $\mathbf{0}$ is the identity element. Let E^0 be the set of all sets in M that contain the null vector. Then, (M3) and (M4) are trivially satisfied, and $(M, \oplus, E^0)$ is an abelian additive MI-monoid of convex compact subsets of $\mathbb{R}^n$.⁷

Example 4.6. Let $\mathbb{R}_{>0}^n$ denote the set of all positive real vectors, and let $\mathbf{a} \cdot \mathbf{b}$ be the product of vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}_{>0}^n$ defined pointwise (analogously to the addition of vectors). For simplicity, we standardly use $\mathbf{a} \cdot \mathbf{b} = \mathbf{ab}$. A subset A of $\mathbb{R}_{>0}^n$ is said to be *multiplication convex* if, for any $\mathbf{a}, \mathbf{b} \in A$ and $\lambda \in [0, 1]$, we have $\mathbf{a}^\lambda \mathbf{b}^{1-\lambda} \in A$. Let N be the set of all multiplication convex compact subsets of $\mathbb{R}_{>0}^n$, and let $\odot$ be the “Minkowski” multiplication defined by

$$A \odot B = \{\mathbf{ab} \mid \mathbf{a} \in A, \mathbf{b} \in B\}.$$

Note that, as in the Minkowski addition, the multiplication of two compact sets is a compact set.⁸ Let us show that $A \odot B$ is multiplication convex. If $\mathbf{x}, \mathbf{y} \in A \odot B$ and $\lambda \in [0, 1]$, then there exist $\mathbf{a}, \mathbf{c} \in A$ and $\mathbf{b}, \mathbf{d} \in B$ such that $\mathbf{x} = \mathbf{ab}$ and $\mathbf{y} = \mathbf{cd}$, and $\mathbf{x}^\lambda \mathbf{y}^{1-\lambda} = (\mathbf{ab})^\lambda (\mathbf{cd})^{1-\lambda} = (\mathbf{a}^\lambda \mathbf{c}^{1-\lambda})(\mathbf{b}^\lambda \mathbf{d}^{1-\lambda})$. Since $\mathbf{a}^\lambda \mathbf{c}^{1-\lambda} \in A$, and $\mathbf{b}^\lambda \mathbf{d}^{1-\lambda} \in B$, then $\mathbf{x}^\lambda \mathbf{y}^{1-\lambda} \in A \odot B$, and therefore $A \odot B$ is product convex. It is easy to see that $\odot$ is commutative and associative, and the set containing only the vector $\mathbf{1} = (1, \dots, 1)$ is the identity element. Let E^1 be the set of all sets in N that contain $\mathbf{1}$. Again, (M3) and (M4) are trivially satisfied, and $(N, \odot, E^1)$ is an abelian multiplicative MI-monoid of multiplication convex compact sets in $\mathbb{R}_{>0}^n$.

One may see that the concept of MI-monoid may serve as a basic algebraic structure for various types of objects and operations defined on them (intervals, sets, fuzzy sets and their arithmetics). This motivates us to develop a theory of MI-monoids and to introduce further types of MI-algebras.

Remark 4.2. It should be noted that the axiomatic system for MI-monoids is too general to precisely specify what is and what is not an appropriate set of pseudoidentities. As a demonstration of curious MI-monoids, we can consider $(G, +, G)$, where $(G, +)$ is an abelian monoid, or $(G, +, E^0)$, where G is the set of all real intervals, $+$ is defined in Example 4.1, and $E^0 = \{[a, a] \mid a \in \mathbb{R}\}$ is the set of all “degenerate” real intervals. In the second example, E^0 represents $\mathbb{R}$ and does not delimit anything, which can be understood to be zero-like elements. Later, it will be clear that many such curious MI-monoids cannot be extended to MI-structures similar to a group or field.

4.2. Homomorphisms and MI-submonoids

Essential to the study of MI-monoids and further algebraic objects stemming from the MI-monoids are the functions that preserve the given algebraic structure. We provide the following definition that naturally generalizes the homomorphism of monoids.

Definition 4.3. Let G and H be MI-monoids. A mapping $f : G \rightarrow H$ is a *homomorphism* of MI-monoids provided

$$(HM1) \quad f(xy) = f(x)f(y) \text{ for all } x, y \in G,$$

$$(HM2) \quad f(e_G) = e_H,$$

$$(HM3) \quad f(a) \in E_H \text{ for all } a \in E_G,$$

where e_G and e_H (E_G and E_H) denote the strong identity (the set of all pseudoidentities) of G and H , respectively. If f is injective, f is said to be a *monomorphism*. If f is surjective, f is said to be an *epimorphism*. If f is bijective,

⁷ Note that the Minkowski addition is used to define the operation of *dilation*, which plays a central role in mathematical morphology (see [39]).

⁸ It immediately follows from the fact that the pointwise multiplication of two vectors is a continuous mapping of $\mathbb{R}^n \times \mathbb{R}^n$ onto $\mathbb{R}^n$, the product of two compact subsets A, B of $\mathbb{R}^n$, i.e., $A \times B$, is a compact set in $\mathbb{R}^n \times \mathbb{R}^n$, and the image under a surjective continuous mapping of a compact set is a compact set.

and $f(E_G) = \{f(a) \mid a \in E_G\} = E_H$, f is said to be an *isomorphism*. In this case, G and H are said to be *isomorphic* (written $G \cong H$). A homomorphism $f : G \rightarrow G$ is called an *endomorphism*, and an isomorphism $f : G \rightarrow G$ is called an *automorphism*.

Clearly, the composition of two homomorphisms (monomorphisms, epimorphisms, etc.) is again a homomorphism (monomorphism, epimorphism, etc.).

Example 4.7. Let $(M, \oplus, E^0)$ and $(N, \odot, E^1)$ be the MI-monoids of convex and multiplication convex compact subsets of $\mathbb{R}^n$ introduced in Examples 4.5 and 4.6, respectively. Then, $f : N \rightarrow M$ given by $f(A) = \{\ln \mathbf{a} \mid \mathbf{a} \in A\}$, where $\ln \mathbf{a}$ is defined pointwise, is an isomorphism of the MI-monoids N and M . In fact, we have $f(\mathbf{ab}) = \ln \mathbf{ab} = \ln \mathbf{a} + \ln \mathbf{b} = f(\mathbf{a}) + f(\mathbf{b})$, $f(\mathbf{1}) = \ln \mathbf{1} = \mathbf{0}$, and if $\mathbf{1} \in A$, then $\mathbf{0} \in f(A)$. Let $A, B \in N$. To check that $f(A)$ is a convex set, let $\mathbf{a}, \mathbf{b} \in f(A)$ and $\lambda \in [0, 1]$. By the definition of f , there are $\mathbf{c}, \mathbf{d} \in A$ such that $f(\mathbf{c}) = \mathbf{a}$ and $f(\mathbf{d}) = \mathbf{b}$. Then,

$$\lambda \mathbf{a} + (1 - \lambda) \mathbf{b} = \lambda \ln \mathbf{c} + (1 - \lambda) \ln \mathbf{d} = \ln(\mathbf{c}^\lambda \mathbf{d}^{1-\lambda}) = f(\mathbf{c}^\lambda \mathbf{d}^{1-\lambda}).$$

Since $\mathbf{c}^\lambda \mathbf{d}^{1-\lambda} \in A$, $\lambda \mathbf{a} + (1 - \lambda) \mathbf{b} \in f(A)$, and $f(A)$ is convex. Further, we have

$$\begin{aligned} f(A \odot B) &= \{\ln(\mathbf{ab}) \mid \mathbf{a} \in A, \mathbf{b} \in B\} \\ &= \{\ln \mathbf{a} + \ln \mathbf{b} \mid \mathbf{a} \in A, \mathbf{b} \in B\} = f(A) \oplus f(B), \end{aligned}$$

whence (HM1) is satisfied. Obviously, $f(\{\mathbf{1}\}) = \{\ln \mathbf{1}\} = \{\mathbf{0}\}$, and if $A \in E^1$, then $\mathbf{0} \in f(A)$, and thus $f(A) \in E^0$. Hence, f is a homomorphism of N to M . If $A \neq B$, then there is $\mathbf{a} \in A \setminus B$. Since the natural logarithm is a bijection of the set of positive real numbers to the set of real numbers, we obtain $f(\mathbf{a}) \notin B$, and f is an injection of N to M . Let $B \in M$. Put $A = \{e^{\mathbf{b}} \mid \mathbf{b} \in B\}$, where again $e^{\mathbf{b}}$ is defined pointwise. Then, $f(A) = \{\ln e^{\mathbf{b}} \mid \mathbf{b} \in B\} = \{\mathbf{b} \mid \mathbf{b} \in B\} = B$, hence f is surjective. Finally, if $f(A) = B$ and $\mathbf{0} \in B$, then $\mathbf{1} \in A$ and $f(E^1) = E^0$.

Let $(G, \circ)$ be a monoid and $H \subseteq G$ be a non-empty subset. If H contains the identity element of G , and the restriction of $\circ$ to H is also a binary operation on H , then $(H, \circ)$ is a submonoid of G . The following definition provides a generalization of the concept of submonoids for MI-monoids.

Definition 4.4. Let $G = (G, \circ, E)$ be an MI-monoid, and let $H \subseteq G$ and $F \subseteq E$ be non-empty subsets. If H contains the strong identity of G and is itself an MI-monoid under the product of G , then $H = (H, \circ, F)$ is said to be an *MI-submonoid* of G . This is denoted by $H \leq G$.

The MI-submonoids $(G, \circ, E)$, $(E, \circ, E)$, and $(\{e\}, \circ, \{e\})$ of an MI-monoid $(G, \circ, E)$ are called *trivial*.

Example 4.8. Real intervals with endpoints in the set of integers equipped with the common interval addition and multiplication, where the pseudoidentities are defined analogously as for real intervals, form an additive and multiplicative MI-submonoid of the MI-monoid of real intervals (see Example 4.1).

Example 4.9. Triangular fuzzy numbers equipped with addition (multiplication), where the pseudoidentities are defined to be symmetrically distributed around zero (one), i.e., $A(x) = A(-x)$ ($A(x) = A(1/x)$) holds for any $x \in \mathbb{R}$, form an additive (multiplicative) MI-submonoid of the MI-monoid of trapezoidal fuzzy numbers (see Example 4.3).

The following proposition shows the necessary and sufficient condition under which subsets of G specify an MI-submonoid of G .

Proposition 4.1. Let $G = (G, \circ, E)$ be an MI-monoid, and let $H \subseteq G$ and $F \subseteq E$ be non-empty subsets. Then, $H \leq G$ if and only if, for all $x, y \in H$ and $a, b \in F$, the following hold:

- (i) $xy \in H$,
- (ii) $e \in F$,
- (iii) $ab \in F$.

Proof. ($\Rightarrow$) Trivial. ($\Leftarrow$) Obviously, the conditions (M1) and (M4) are satisfied, since the operation on H is the restriction of the operation on G . The conditions (M2) and (M3) are equivalent to (ii) and (iii), respectively. $\square$

An MI-submonoid H of an MI-monoid G is defined by a suitable pair of subsets $H \subseteq G$ and $F \subseteq E$. If we set $F = H \cap E$, we obtain a special case of MI-submonoid induced only by a subset H of a set G .

Definition 4.5. An MI-submonoid $(H, \circ, F)$ of an MI-monoid $(G, \circ, E)$ is said to be *canonical* if $F = H \cap E$.

The following proposition shows the necessary and sufficient condition for $H \subseteq G$ to define a canonical MI-submonoid, which is simpler than the condition provided in Proposition 4.1. Moreover, its form is analogous to the standard one for monoids (see [16]).

Proposition 4.2. Let $G = (G, \circ, E)$ be an MI-monoid, and let $H \subseteq G$ be a non-empty subset. Then, H is a canonical MI-submonoid of G if and only if $e \in H$ and $xy \in H$ for all $x, y \in H$.

Proof. ($\Rightarrow$) Trivial. ($\Leftarrow$) We prove that (iii) of Proposition 4.1 is true. If $a, b \in H \cap E$, then $ab \in H$ and $ab \in E$. Hence, we obtain $ab \in H \cap E$, and (iii) is satisfied. $\square$

One can see that the MI-submonoids of the real intervals from Example 4.8 are canonical, but the MI-submonoids of the trapezoidal fuzzy numbers from Example 4.9 are not canonical, as the pseudoidentities are restricted here to the triangular fuzzy sets symmetrically distributed around the strong identity.

Example 4.10. Let $(\mathcal{F}_S(\mathbb{R}), +, E_S^0)$ be the above introduced abelian additive MI-monoid of fuzzy numbers that are extensional w.r.t. $*$ -similarity relations from a closed system $\mathcal{S}$. Let $\mathcal{T} \subseteq \mathcal{S}$ be a closed subsystem of $\mathcal{S}$. Using the previous proposition, one may simply check that $(\mathcal{F}_T(\mathbb{R}), +, E_T^0)$, where clearly $\mathcal{F}_T(\mathbb{R}) \subseteq \mathcal{F}_S(\mathbb{R})$ and $E_T^0 = E_S^0 \cap \mathcal{F}_T(\mathbb{R})$, is a canonical MI-submonoid of $\mathcal{F}_S(\mathbb{R})$.

Example 4.11. Let $(H, \cdot, E^1)$ be the multiplicative MI-monoid, and let E^0 be the set of pseudoidentities of the additive MI-monoid $(H, +, E^0)$ of fuzzy numbers from Example 4.2. Then, $(H \setminus E^0, \cdot, E^1 \setminus E^0)$ is a canonical MI-submonoid (of “non-zero fuzzy numbers”) of $(H, \cdot, E^1)$.

Example 4.12. Let $\mathbb{R}_{\geq 0}^n$ denote the set of all non-negative real vectors, and let $M_{\geq 0}$ be the set of all (non-negative) convex compact subsets of $\mathbb{R}_{\geq 0}^n$. In Example 4.5, we defined an abelian MI-monoid $M = (M, \oplus, E^0)$, where $\oplus$ is the Minkowski addition. Since $\mathbf{0} \in M_{\geq 0}$ and $A \oplus B \in M_{\geq 0}$ for any $A, B \in M_{\geq 0}$, then, by the previous proposition, $(M_{\geq 0}, \oplus, E^0 \cap M_{\geq 0})$ is a canonical MI-submonoid of M .

If we deal with pseudoidentities, then it is reasonable to introduce a relation $\sim$ on G that is defined as follows. Two elements $x, y \in G$ are in a relation $\sim$, denoted by $x \sim y$, if there exist $a, b \in E$ such that $ax = by$ (cf. [4,27,28]). Note that $x \sim y$ if and only if $xa = yb$, which follows from (M4).

Proposition 4.3. The relation $\sim$ is a congruence on G .

Proof. Obviously, $x \sim x$ and $x \sim y = y \sim x$ are trivially satisfied. Let $x \sim y$ and $y \sim z$. Then, $ax = by$ and $cy = dz$ for some $a, b, c, d \in E$. Hence, we have

$$\begin{aligned}(ca)x &= c(ax) = c(by) = (cb)y \\ (bd)z &= b(dz) = b(cy) = (bc)y = (cb)y,\end{aligned}$$

where (M1) and (M4) are applied, and thus $(ca)x = (bd)z$. Since $ca, bd \in E$, we obtain $x \sim z$ and $\sim$ is an equivalence relation on G . Now, let us prove that $\sim$ is compatible with the operation. If $x \sim x'$ and $y \sim y'$, then $ax = bx'$ and $cy = dy'$ for suitable $a, b, c, d \in E$. Hence, $(ac)(xy) = (ax)(cy) = (bx')(dy') = (bd)(x'y')$. Since $ac, bd \in E$, we obtain $xy \sim x'y'$. $\square$

5. MI-pregroups and MI-groups

5.1. Main definitions and examples

The concept of the MI-monoid is just a first step that allows us to deal with more identity-like elements, but the next natural step is to introduce inverse elements, and thus a structure that naturally generalizes group structure. In this section, we aim for this goal and continue in the development of a theory of MI-algebras. We define an important extension of a group to an MI-group. We stress the word “important” because fuzzy numbers (as well as, e.g., real intervals, convex compact sets) equipped with an operation do not retain the group structure, in contrast to the monoid structure. Moreover, the groups and their theory play a prominent role among algebraic structures, and thus the same role for MI-groups is expected.

Because the definition of an MI-group is a bit complicated for imitating the important properties of groups, we will start with the definition of the inversion on an MI-monoid. Recall that, in any group, for each $x \in G$, there exists an inverse $x^{-1} \in G$ such that $xx^{-1} = x^{-1}x = e$. A simple consequence of the group definition is that each element of a group possesses precisely one inverse element, $(xy)^{-1} = y^{-1}x^{-1}$, and $(x^{-1})^{-1} = x$. Let $(G, \circ)$ be a group, and define a dual group as $G^{\text{op}} = (G, \circ^{\text{op}})$, where $x \circ^{\text{op}} y = y \circ x$. Then, it is easy to see that $f(x) = x^{-1}$ defines an isomorphism $f : G \rightarrow G^{\text{op}}$ satisfying the following properties:

$$f(x) \circ x = x \circ f(x) = e \quad \text{and} \quad f(f(x)) = x.$$

Clearly, it is sufficient to consider $x \circ f(x) = e$ whenever the group G is abelian. Moreover, each monoid G is a group if and only if there exists an isomorphism $f : G \rightarrow G^{\text{op}}$ with the properties mentioned above. This isomorphism f is unique.⁹

Definition 5.1. Let $G = (G, \circ, E)$ be an MI-monoid. A *dual* MI-monoid to G is $G^{\text{op}} = (G, \circ^{\text{op}}, E)$, where $\circ^{\text{op}}$ is given by $x \circ^{\text{op}} y = y \circ x$.

Remark 5.1. Clearly, $G = G^{\text{op}}$ whenever G is an abelian MI-monoid.

Definition 5.2. An MI-monoidal isomorphism f of G onto G^{op} that satisfies

- (G1) $f(x) \circ x \in E$,
- (G2) $f(x) \circ x = x \circ f(x)$,
- (G3) $f(f(x)) = x$

for any $x \in G$ is called an *f-inversion* (inversion, for short) in G .

Again, we use $f(x)x = f(x) \circ x$. One can see that (G1) is the only difference from the isomorphism that defines the inverse elements of a group. Nevertheless, as we have mentioned above, the pseudoidentity elements concentrate the fuzziness in the elements, and it would be counterintuitive to require the satisfaction of the law $f(x)x = e$. However, we naturally obtain $f(x)x \sim e$, which expresses the fact that $f(x)x$ differs from e only up to a pseudoidentity.

One might notice that $f(f(x)x) = f(x)x$. Hence, there are elements in G for which $f(x) = x$. This motivates us to introduce the following type of elements.

Definition 5.3. Let G be an MI-monoid, and let f be an inversion in G . An element $x \in G$ with $f(x) = x$ is said to be *f-symmetric* (*symmetric, for short*) in G .

⁹ In fact, if $f, g : G \rightarrow G^{\text{op}}$ are two isomorphisms satisfying the mentioned properties, then, for any $x \in G$, $f(x) \circ x = g(x) \circ x = e$ implies $f(x) = (f(x) \circ x) \circ f(x) = (g(x) \circ x) \circ f(x) = g(x) \circ (x \circ f(x)) = g(x)$, whence $f(x) = g(x)$ for any $x \in G$, and thus f is unique. Note that the property $f(f(x)) = x$ is a consequence of f being an isomorphism that satisfies $f(x) \circ x = x \circ f(x) = e$.

Proposition 5.1. Let $G = (G, \circ, E)$ be an MI-monoid, and let f be an inversion in G . Let P_G^f be the least submonoid of $(G, \circ)$ containing the set $\{xf(x) \mid x \in G\}$. Then, P_G^f is an abelian submonoid of f -symmetric elements of the monoid $(G, \circ)$.

Proof. As mentioned above, $xf(x)$ is an f -symmetric element in G for any $x \in G$. Since $X = \{xf(x) \mid x \in G\} \subseteq E$, and E is an abelian submonoid of G , $P_G^f \subseteq E$; therefore, P_G^f is an abelian submonoid of G . Note that $e \in P_G^f$. Obviously, $P_G^f = \{a_1 a_2 \cdots a_n \mid a_i \in X, n \in \mathbb{N}\}$, where $\mathbb{N}$ denotes the set of all natural numbers. Let $b \in P_G^f$, i.e., $b = x_1 f(x_1) \cdots x_n f(x_n)$ for some $x_1, \dots, x_n \in G$. Using (M4) and the fact that $x_i f(x_i)$, $i = 1, \dots, n$, are f -symmetric in G , we have $f(b) = f(x_1 f(x_1)) \cdots f(x_n f(x_n)) = x_1 f(x_1) \cdots x_n f(x_n) = b$. Hence, b is an f -symmetric element in G . $\square$

The set P_G^f could be interpreted as the least admissible set of pseudoidentities containing “active pseudoidentities” in G , where an active pseudoidentity means the product of an element and its inversion in G . For example, the identity element e is an active pseudoidentity in each MI-monoid.

In contrast to the unique inverse that exists for each element in the groups, we can suppose the existence of more than one inversion in MI-monoids. The following proposition shows the relation between two inversions in an MI-monoid.

Proposition 5.2. Let $G = (G, \circ, E)$ be an MI-monoid. If f and g are inversions in G , then, for any $x \in G$, there exist $s \in P_G^f$ and $t \in P_G^g$ such that $f(x)t = g(x)s$, i.e., $f(x) \sim g(x)$.

Proof. Let $x, y \in G$, and let f, g be two inversions in G . Put $s = xf(x)$ and $t = xg(x)$. By the definition, we have $s \in P_G^f$ and $t \in P_G^g$. Then, $f(x)t = f(x)xg(x) = sg(x) = g(x)s$, where the last equality follows from (M4). $\square$

Thus, one can see that arbitrary two inversions defined on an MI-monoid are different up to symmetric pseudoidentities. If $P_G^f = P_G^g = \{e\}$, then f and g coincide, and there is a unique inverse in G . Our analysis of arithmetics based on lattice operations shows that MI-monoids equipped with inversions do not satisfy the cancellation law in general. This fact is a significant difference from groups, where the cancellation law is a consequence of the inversion. To cope with this problem, we define an MI structure equipped with inversion that does not fulfill the cancellation law. This structure will be called an MI-pregroup.

Definition 5.4. A quadruplet $G = (G, \circ, E, f)$ is said to be an MI-pregroup if $(G, \circ, E)$ is an MI-monoid and f is an inversion in $(G, \circ, E)$. An MI-pregroup G is said to be good if G is a good MI-monoid.

Example 5.1. The algebraic structure for the summation of extensional fuzzy numbers $(\mathcal{F}_S(\mathbb{R}), +, E_S^0, f^0)$ with inverse elements defined as $f^0(x_S) = (-x)_S$ is a good abelian additive MI-pregroup. The algebraic structure for the multiplication of extensional fuzzy numbers $(\mathcal{F}_S(\mathbb{R}) \setminus E_S^0, \cdot, E_S^1, f^1)$ with inverse elements defined as $f^1(x_S) = (1/x)_S$ is a good abelian multiplicative MI-pregroup. In both examples, all pseudoidentity elements are symmetric, and, moreover, $E_S^0 = P_{\mathcal{F}_S(\mathbb{R})}^{f^0}$, and similarly for multiplication.

The following example provides an interesting relationship between the fuzzy subgroups of a group G and MI-pregroups.

Example 5.2. Let $*$ be a left-continuous t-norm, and recall that a $\star$ -fuzzy subgroup of a group $(G, \circ)$ is a fuzzy set $H : G \rightarrow [0, 1]$ satisfying the following axioms [34,35,38]:

- (i) $H(x) * H(y) \leq H(x \circ y)$ for any $x, y \in G$,
- (ii) $H(x) \leq H(x^{-1})$ for any $x \in G$,
- (iii) $H(e) = 1$,

where e denotes the identity element of G . Define a subset $\mathbb{H}$ of $G \times (0, 1]$, a binary operation $\bullet$ and an inversion f on $\mathbb{H}$ as follows:

$$\begin{aligned}(x, \alpha) &\in \mathbb{H} \quad \text{iff } \alpha \leq H(x), \\ (x, \alpha) \bullet (y, \beta) &= (x \circ y, \alpha * \beta), \\ f(x, \alpha) &= (x^{-1}, \alpha).\end{aligned}$$

Put $\mathbb{E} = \{(e, \alpha) \mid \alpha \in (0, 1]\}$. One can check that if there are no zero divisors with respect to $*$, then $(\mathbb{H}, \bullet, \mathbb{E}, f)$ is a good MI-pregroup.

As mentioned above, the cancellation law need not be always fulfilled, which led us to introduce MI-pregroups, but it seems to be an important property in the further analysis of MI-monoids with inversions [14]. Therefore, the next natural step is the definition of an MI-group.

Definition 5.5. A quadruplet $G = (G, \circ, E, f)$ is said to be an *MI-group* if $(G, \circ, E, f)$ is an MI-pregroup satisfying the following restricted cancellation law, i.e.,

$$(G4) \text{ if } xs = ys \text{ for } x, y \in G \text{ and } s \in P_G^f \text{ then } x = y.$$

An MI-group G is said to be *good* if G is a good MI-pregroup.

One could be surprised that we use the cancellation law restricted to the pseudoidentities from P_G^f . The following proposition shows that the left and right cancellation law without any restriction is also true in MI-groups.

Proposition 5.3. Let $G = (G, \circ, E, f)$ be an MI-group. Then, the (left, right) cancellation law is satisfied in G .

Proof. Let us suppose that the restricted cancellation law holds in G and $x, y, z \in G$ such that $zx = zy$. Then, $(f(z)z)x = (f(z)z)y$, where $f(z)z \in P_G^f$. Using (M4), we obtain $x(f(z)z) = y(f(z)z)$, and therefore $x = y$ by the restricted cancellation law. The case $xz = yz$ can be proved using analogous arguments. $\square$

It is easy to see that neither $(\mathcal{F}_S(\mathbb{R}), +, E_S^0, f^0)$ nor $(\mathcal{F}_S(\mathbb{R}) \setminus E_S^0, \cdot, E_S^1, f^1)$ from Example 5.1 are MI-groups, since the union $\cup$ does not fulfill the cancellation law, i.e., there are $*$ -similarity relations R, S, T in $\mathcal{S}$ such that $R \cup S = R \cup T$, but $S \neq T$, in general. However, we can provide many examples of MI-groups based on the MI-structures introduced in the previous sections.

Example 5.3. Let $\mathcal{R}$ be the closed system of $*$ -similarity relations from Example 3.3. The algebraic structure for the summation of extensional fuzzy numbers $(\mathcal{F}_{\mathcal{R}}(\mathbb{R}), \boxplus, E_{\mathcal{R}}^0, f^0)$ with inverse elements defined as $f^0(x_R) = (-x)_R$ is a good abelian additive MI-group. The algebraic structure for the multiplication of extensional fuzzy numbers $(\mathcal{F}_{\mathcal{R}}(\mathbb{R}) \setminus E_{\mathcal{R}}^0, \boxtimes, E_{\mathcal{R}}^1, f^1)$ with inverse elements defined as $f^1(x_R) = (1/x)_R$ is a good abelian multiplicative MI-group.

In both examples, all pseudoidentity elements are symmetric, and, moreover, $E_{\mathcal{R}}^0 = P_{\mathcal{F}_{\mathcal{R}}(\mathbb{R})}^{f^0}$, and similarly for the multiplication.

Example 5.4. Let us consider examples of structures with real intervals from Example 4.1. For the additive MI-monoid, we may define the inverse elements as $f^0([a, b]) = [-b, -a]$, and we obtain an abelian MI-group that is not good. For the multiplicative structure, we first omit E^0 elements from G to obtain the reduced structure $(G \setminus E^0, \cdot, E^1 \setminus E^0)$. Then, we may define inverse elements as $f^1([a, b]) = [1/b, 1/a]$, and we obtain an abelian MI-group that is not good.

It should be noted that, in these examples, not all pseudoidentities are symmetric. Indeed, check, e.g., $[-3, 2] \in E^0$ and its inversion in the additive structure $[-2, 3] \in E^0$ but $[-3, 2] \neq [-2, 3]$.

Example 5.5. Let H be the abelian additive MI-monoid of fuzzy numbers from Example 4.2, and define $f^0(g)(\alpha) = -g(\alpha)$ for any $g \in H$ and $\alpha \in [0, 1]$, where $-g(\alpha) = [-g(\alpha)^+, -g(\alpha)^-]$ for $g(\alpha) = [g(\alpha)^-, g(\alpha)^+]$. It is easy to see that $f^0(g) \in H$, and f^0 is an inversion in H . Hence, $(H, +, E^0, f^0)$ is an abelian additive MI-group.

Let $(H \setminus E^0, \cdot, E^1 \setminus E^0)$ be the multiplicative MI-monoid from Example 4.11, and define $f^1(g)(\alpha) = g(\alpha)^{-1}$ for any $g \in H \setminus E^0$ and $\alpha \in [0, 1]$, where $g(\alpha)^{-1} = [1/g(\alpha)^+, 1/g(\alpha)^-]$ for $g(\alpha) = [g(\alpha)^-, g(\alpha)^+]$. Then, $(H \setminus E^0, \cdot, E^1 \setminus E^0, f^1)$ is an abelian multiplicative MI-group.

Example 5.6. Let L be the additive MI-monoid from Example 4.4, and define $f^0(u) = (-u_i^+, -\delta u_i^+, -u_i^-, -\delta u_i^-)_{i=0, \dots, N}$ for any $u \in L$. Then, $(L, +, E^0, f^0)$ is an abelian additive MI-group. One may check that $(L \setminus E^0, \cdot, E^1 \setminus E^0)$ is an MI-submonoid of $(L, \cdot, E^1)$. If we define

$$f^1(u) = (1/u_i^+, -1/(u_i^+)^2 \delta u_i^+, 1/u_i^-, -1/(u_i^-)^2 \delta u_i^-)_{i=0, \dots, N}$$

for any $u \in L \setminus E^0$, then $(L \setminus E^0, \cdot, E^1 \setminus E^0, f^1)$ is an abelian multiplicative MI-group.

Example 5.7. Let M be the additive MI-monoid from Example 4.5, and define $f^0(A) = -A = \{-a \mid a \in A\}$ for any $A \in M$. Then, $(M, \oplus, E^0, f^0)$ is an abelian additive MI-group of convex compact subsets of $\mathbb{R}^n$.¹⁰ Analogously, one may define an abelian multiplicative MI-group N of multiplication convex compact subsets of $\mathbb{R}^n$.

Example 5.8. Let $\mathbb{H} = (\mathbb{H}, \bullet, \mathbb{E}, f)$ be a good MI-pregroup defined in Example 5.2 from a $*$ -fuzzy subgroup of a group G . If $*$ satisfies the cancellation law, then $\mathbb{H}$ is an MI-group.

Convention. Where no confusion can arise, for the sake of readability of the text, we will omit f in $(G, \circ, E, f)$ and standardly write x^{-1} instead of $f(x)$ for the inversion in G . Moreover, we will use P_G instead of P_G^f to denote the least admissible set of pseudoidentities containing “active pseudoidentities” in G .

5.2. Homomorphisms and MI-subgroups

In this section, we restrict ourselves to MI-groups. Nevertheless, the concepts of homomorphism and subgroup can be analogously introduced for MI-pregroups. A natural generalization of a homomorphism of groups under which we can study the important properties of MI-groups is as follows.

Definition 5.6. Let G and H be MI-groups. A mapping $f : G \rightarrow H$ is a *homomorphism of MI-groups* if

- (HG1) $f(xy) = f(x)f(y)$ for all $x, y \in G$,
- (HG2) $f(x) \in E_H$ for all $x \in E_G$,
- (HG3) $f(x^{-1}) = (f(x))^{-1}$ for all $x \in G$,

where E_G and E_H denote the sets of pseudoidentities of G and H , respectively.

The concepts, among others, of monomorphism, epimorphism, and isomorphism of MI-groups have the same meanings as those for MI-monoids. Moreover, if $f : G \rightarrow H$ and $g : H \rightarrow P$, then $g \circ f$ is a homomorphism of G to P . In fact, if $x \in G$, then $f(x^{-1}) = f(x)^{-1}$ and $g(f(x)^{-1}) = g(f(x))^{-1}$. Hence, we obtain $g(f(x^{-1})) = g(f(x))^{-1}$, and (HG3) is satisfied. Similarly, one can prove the remaining conditions. In some cases, it is useful to assume an additional condition that is trivially satisfied for groups. Namely, all active pseudoidentities in G are mapped onto all active pseudoidentities in H . The precise definition is as follows.

Definition 5.7. A homomorphism $f : G \rightarrow H$ of MI-groups satisfying

- (HG4) $f(P_G) = P_H$

is said to be a *strong* homomorphism of MI-groups.

¹⁰ Note that $-A$ is called the *reflection* of A and that it is used to define the *erosion* in the mathematical morphology (see, e.g., [39]). The fulfillment of the cancellation law for the Minkowski addition of convex compact subsets of $\mathbb{R}^n$ is a well-known fact (see, e.g., [29,31]).

Obviously, the composition of strong homomorphisms is a strong homomorphism, but not all homomorphisms are strong. The following theorem shows a sufficient condition for a strong homomorphism.

Theorem 5.4. *Each epimorphism is strong.*

Proof. Let $f : G \rightarrow H$ be an epimorphism. According to Theorem 5.5, it is sufficient to show that $P_H \subseteq f(P_G)$. Let $s \in P_H$. By the definition of P_H , $s = z_1 z_1^{-1} \cdots z_n z_n^{-1}$ for some $z_1, \dots, z_n \in H$. Since f is an epimorphism, we have

$$f(y_1 y_1^{-1} \cdots y_n y_n^{-1}) = f(y_1) f(y_1)^{-1} \cdots f(y_n) f(y_n)^{-1} = z_1 z_1^{-1} \cdots z_n z_n^{-1} = s$$

for suitable $y_1, \dots, y_n \in G$. Since $y_1 y_1^{-1} \cdots y_n y_n^{-1} \in P_G$, we obtain the desired inclusion. $\square$

Example 5.9. Each inversion of an MI-group is a strong MI-group automorphism.

Example 5.10. The isomorphism f of the MI-monoids N and M defined in Example 4.7 is also a strong isomorphism of the MI-groups N and M (see Example 5.7).

The following theorem shows how the identity element e_G and the set P_G of (active) pseudoidentity elements are preserved with respect to an MI-group homomorphism.

Theorem 5.5. *Let $f : G \rightarrow H$ be a homomorphism of MI-groups. Then,*

- (i) $f(P_G) \subseteq P_H$.
- (ii) $f(e_G) = e_H$.

Proof. (i) Let $a \in P_G$. Then $a = x_1 x_1^{-1} \cdots x_n x_n^{-1}$, and thus

$$f(a) = f(x_1 x_1^{-1} \cdots x_n x_n^{-1}) = f(x_1) f(x_1)^{-1} \cdots f(x_n) f(x_n)^{-1} \in P_H.$$

(ii) If $x \in G$, then $f(e_G x) = f(e_G) f(x) = e_H f(x)$. By the cancellation law, which holds in H , we obtain $f(e_G) = e_H$. $\square$

A very important concept in group theory is the kernel of a homomorphism. For MI-groups, we propose the following definition.

Definition 5.8. Let $f : G \rightarrow H$ be a homomorphism of MI-groups. The *kernel* of f is the set $\text{Ker } f = \{x \in G \mid f(x) \in P_H\}$.

The following theorem belongs to the folklore in group theory and specifies the monomorphism and isomorphism of MI-groups in terms of the kernels and compositions of two homomorphisms, respectively. To provide a sound formulation of this theorem, we require a strong homomorphism. Nevertheless, this assumption is needed only in the proof of one implication, as discussed below in Remark 5.2. Denote by id_G the identity homomorphism of MI-groups.

Theorem 5.6. *Let $f : G \rightarrow H$ be a strong homomorphism of MI-groups. Then,*

- (i) f is a monomorphism if and only if $\text{Ker } f = P_G$ and f restricted to P_G is an injective mapping,
- (ii) f is an isomorphism if and only if there is a homomorphism $f^{-1} : H \rightarrow G$ such that $f f^{-1} = \text{id}_H$ and $f^{-1} f = \text{id}_G$.

Proof. (i) $(\Rightarrow)$ Let f be a strong monomorphism. Trivially, f restricted to P_G is a bijective mapping onto P_H , which immediately implies the equality $\text{Ker } f = P_G$.

$(\Leftarrow)$ Let $\text{Ker } f = P_G$, and let f restricted to P_G be an injective mapping. If $f(x) = f(y)$ for some $x, y \in G$, then $f(x y^{-1}) = f(x) f(y)^{-1} = f(y) f(y)^{-1} = f(y y^{-1}) \in P_H$. Since $\text{Ker } f = P_G$, both $x y^{-1}$ and $y y^{-1}$ are from P_G . Due

to the injectivity of f on P_G , they must equal $xy^{-1} = yy^{-1}$. By the cancellation law, we obtain $x = y$, and thus f is a monomorphism.

(ii) ($\Rightarrow$) Let $f : G \rightarrow H$ be an isomorphism, and let $f^{-1} : H \rightarrow G$ be defined in such a way that $ff^{-1} = id_H$ and $f^{-1}f = id_G$ simultaneously hold. Note that f^{-1} is a bijective mapping. We will prove that f^{-1} is a homomorphism. If $x, y \in H$, then $f(f^{-1}(xy)) = xy$ and $f(f^{-1}(x)f^{-1}(y)) = f(f^{-1}(x))f(f^{-1}(y)) = xy$. Hence, $f(f^{-1}(xy)) = f(f^{-1}(x)f^{-1}(y))$. Since f is bijective, $f^{-1}(xy) = f^{-1}(x)f^{-1}(y)$, and (HG1) is satisfied. Since f is an isomorphism, then $f(F) = E$, and therefore, $f^{-1}(E) = F$, and (HG2) is true. If $x \in H$, then $f(f^{-1}(x^{-1})) = x^{-1}$ and $f((f^{-1}(x))^{-1}) = (ff^{-1}(x))^{-1} = x^{-1}$. Hence, $f(f^{-1}(x^{-1})) = f((f^{-1}(x))^{-1})$. Since f is bijective, then $f^{-1}(x^{-1}) = (f^{-1}(x))^{-1}$, and (HG3) holds.

($\Leftarrow$) If $f^{-1} : H \rightarrow G$ is a homomorphism such that $ff^{-1} = id_H$ and $f^{-1}f = id_G$, then f is a bijection. Now, by the definition of isomorphism of MI-groups, it is sufficient to prove that $f(E_G) = E_H$. Using (HG2), we have $f(E_G) \subseteq E_H$, and simultaneously, $f^{-1}(E_H) \subseteq E_G$. Therefore, $f(E_G) \subseteq E_H = ff^{-1}(E_H) \subseteq f(E_G)$, which implies $f(E_G) = E_H$, and thus f is an isomorphism. $\square$

Remark 5.2. Notice that the property of being a strong homomorphism was used only in the proof that the condition “ $\text{Ker } f = P_G$ and f restricted to P_G is an injective mapping” is a necessary condition for f to be a monomorphism. Thus, if $\text{Ker } f = P_G$ and f restricted to P_G is an injective mapping, then f must necessarily be a monomorphism even if f is not strong! One can simply show that if P_G consists of all symmetric elements, then (i) is true without the assumption of strong homomorphism.

The concept of an MI-subgroup of an MI-group is naturally defined as follows.

Definition 5.9. Let $G = (G, \circ, E)$ be an MI-group, and let $H \subseteq G$ and $F \subseteq E$ be non-empty sets. If $H = (H, \circ, F)$ is itself an MI-group under the product and the inversion of G , then H is said to be an *MI-subgroup* of G . This is denoted by $H \leq G$.

Example 5.11. Let G be an MI-group. Clearly, G and $(\{e\}, \circ, \{e\}, id_{\{e\}})$ are the trivial MI-subgroups of G . Moreover, $(P_G, \circ, P_G, id_{P_G})$ is an MI-subgroup. This MI-subgroup can be understood as a generalization of the trivial group $(\{e\}, \circ, \{e\})$ because the “active” pseudoidentities perfectly simulate the properties of the identity elements as one could notice above.

The following proposition shows the necessary and sufficient condition under which subsets of G and F specify an MI-subgroup of G .

Proposition 5.7. Let $H \subseteq G$ and $F \subseteq E$ be non-empty subsets of an MI-subgroup of G . Then, $H \leq G$ if and only if, for all $x, y \in H$ and $a, b \in F$, the following hold:

- (i) $xy^{-1} \in H$,
- (ii) $e \in F$,
- (iii) $ab \in F$,
- (iv) $a^{-1} \in F$,
- (v) $x^{-1}x \in F$.

Proof. ($\Rightarrow$) Trivial. ($\Leftarrow$) Since $e \in F$, then, by (i), $ex^{-1} = x^{-1} \in H$ for any $x \in H$. Moreover, $(x^{-1})^{-1} = x$ is preserved by the restriction of the inversion to H . If $x, y \in H$, then $y^{-1} \in H$ and $x(y^{-1})^{-1} = xy \in H$. Hence, and from (ii)–(iv), we obtain that H is an MI-submonoid of G by Proposition 4.1. Let $f : H \rightarrow H^{\text{op}}$ be given by $f(x) = x^{-1}$. We will prove that f is the inversion in H .¹¹ Clearly, the satisfaction of (HM1) and (HM2) by f immediately follows from the fact that $f(x) = x^{-1}$ is the restriction of the inversion in G . Moreover, due to (iv),

¹¹ Note that, although the restriction of an inversion to an MI-submonoid H preserves many of the properties of the original inversion, it need not be an inversion in H in general.

f also satisfies (HM3), and thus f is a monoidal homomorphism. The injectivity of f is preserved by the inversion in G , and the surjectivity is a simple consequence of $f(f(x)) = (x^{-1})^{-1} = x$. Using (iv), we simply obtain $f(F) = F$, and thus f is a monoidal isomorphism. The satisfaction of (G2) and (G3) by f is preserved by the inversion in G . By (v), f satisfies (G1), and f is an inversion in H . Thus, the restriction of the inversion in G to H is again an inversion. If $x, y, z \in H$ such that $xz = yz$, then this equality is also true in G . Hence, $x = y$, which implies (G4), and H is an MI-subgroup of G . $\square$

Analogously to the definition of a canonical MI-submonoid, we can define a canonical MI-subgroup, which is determined by only one subset.

Definition 5.10. An MI-subgroup $(H, \circ, F)$ of an MI-group $(G, \circ, E)$ is said to be *canonical* if $F = H \cap E$.

The following proposition shows the necessary and sufficient condition for $H \subseteq G$ to define a canonical MI-subgroup, which is significantly simpler than the condition provided in Proposition 5.7 (cf., [16]).

Proposition 5.8. Let $G = (G, \circ, E)$ be an MI-group, and let $H \subseteq G$ be a non-empty subset. Then, $H = (H, \circ, H \cap E)$ is an MI-subgroup of G if and only if $e \in H$ and $xy^{-1} \in H$ for all $x, y \in H$.

Proof. $(\Rightarrow)$ Trivial. $(\Leftarrow)$ We prove (ii)–(v) of Proposition 5.7. If $x \in H$, then $x^{-1} = ex^{-1} \in H$. Hence, $xy = x(y^{-1})^{-1} \in H$ for any $x, y \in H$. Set $F = H \cap E$. Obviously, $e \in F$, and (ii) is true. If $a, b \in F$, then $ab \in H$, and also $ab \in E$. Hence, $ab \in F$, and (iii) is proved. If $a \in F$, then $a \in E$, and $a^{-1} \in E$. Since $a^{-1} \in H$ as well, then $a^{-1} \in F$, and (iv) is proved. If $x \in H$, then $xx^{-1} \in H$, and also $xx^{-1} \in E$. Hence, $xx^{-1} \in F$, and (v) is satisfied. $\square$

We show several examples of MI-subgroups of the MI-groups introduced in the previous subsection.

Example 5.12. Let $(H \setminus E^0, \cdot, E^1 \setminus E^0, f^1)$ be the abelian multiplicative MI-group of the non-zero fuzzy numbers defined in Example 5.5. Define the set of all positive fuzzy numbers as

$$H_{>0} = \{g \mid g \in H \setminus E^0 \text{ \& } g(0) \subseteq (0, \infty)\}.$$

Then, one may simply check that $(H_{>0}, \cdot, P_{H_{>0}}, f^1)$, where f^1 denotes the restriction of the inversion in $H \setminus E^0$ to $H_{>0}$, is an MI-subgroup of all positive fuzzy numbers of $H \setminus E^0$, which is not canonical.

Example 5.13. Let $M_B = \{B(x, r) \mid x \in \mathbb{R}^n, r \in \mathbb{R}^+\}$ where $B(x, r) = \{y \in \mathbb{R}^n \mid |x - y| \leq r\}$. In Example 5.7, we defined the abelian MI-group $(M, \oplus, E^0, f^0)$. It is easy to see that $(M_B, \oplus, E^0 \cap M_B, f^0)$ is a canonical MI-subgroup of M .

6. MI-prefields and MI-fields

One original motivation of our analysis of the properties of arithmetical operations with fuzzy numbers was to develop working arithmetics of fuzzy numbers. Our idea was to introduce a structure of fuzzy numbers with appropriate arithmetical operations that is very close to the field of real numbers. The essential step in this concept was a generalization of the group structure leading to the idea of an MI-(pre)group. Now, we are ready to introduce a generalization of the field structure.

Definition 6.1. An algebra $(G, +, \cdot, E^0, E^1, f^0, f^1)$ is said to be an *MI-(pre)field* if

- (F1) $(G, +, E^0, f^0)$ is an abelian additive MI-(pre)group,
- (F2) $(G \setminus E^0, \cdot, E^1 \setminus E^0, f^1)$ is an abelian multiplicative MI-(pre)group,
- (F3) for any $x, y, z \in G$, the following *distributive law*

$$x(y + z) = xy + xz$$

holds.

One can see that the definition of an MI-(pre)field is nearly the same as the definition of a field, only that the abelian additive and multiplicative groups are replaced by the corresponding abelian MI-(pre)groups. Analogously, the multiplicative MI-(pre)group is defined on the support of the additive group without zero-like elements, as the set E^0 specifies the elements for which we cannot define the multiplicative inversion.

Recall that in the case of the MI-prefield, we cannot suppose the satisfaction of the cancellation law. We show some examples of MI-prefields and MI-fields.

Example 6.1. For any closed system of $*$ -similarity relations $\mathcal{S}$, the set of extensional fuzzy numbers equipped with $+$ and $\cdot$ defined in Theorem 3.5 (see also Example 5.3) forms a structure

$$(\mathcal{F}_{\mathcal{S}}(\mathbb{R}), +, \cdot, E_{\mathcal{S}}^0, E_{\mathcal{S}}^1, f^0, f^1)$$

that is an MI-prefield.

Example 6.2. For the closed system of $*$ -similarity relations $\mathcal{R}$, the set of extensional fuzzy numbers equipped with $\boxplus$ and $\boxdot$ introduced in Example 3.5 (see also Example 5.3) forms a structure

$$(\mathcal{F}_{\mathcal{R}}(\mathbb{R}), \boxplus, \boxdot, E_{\mathcal{R}}^0, E_{\mathcal{R}}^1, f^0, f^1)$$

that is an MI-field. Unfortunately, as we discussed in Example 3.5, the propagation of uncertainty modeled by this structure is not compatible with human intuition.

Example 6.3. Let $F = (F, +, \cdot)$ be a field. Then, it is not difficult to show that the 7-tuple

$$(F^2, \oplus, \odot, E^0, E^1, f^0, f^1),$$

where $F^2 = F \times F$ is the Cartesian product of the set F ,

$$(a, b) \oplus (c, d) = (a + c, b + d),$$

$$(a, b) \odot (c, d) = (ac, ad + bc),$$

$E^0 = \{(0, b) \mid b \in F\}$, $E^1 = \{(1, 0)\}$, $f^0(a, b) = (-a, -b)$, and $f^1(a, b) = (a^{-1}, ba^{-2})$, is an MI-field. One can notice that the only difference between a field and the MI-field defined here is the set E^0 . However, E^0 specifies the smallest set of elements for which there exist no inverse elements with respect to $\odot$, and therefore this set must be removed from the multiplicative structure.

As mentioned in the Introduction, generally, the distributivity law is not satisfied for the arithmetical operations for intervals and fuzzy numbers. Nevertheless, it is interesting that for any x, y, z , we may find a pseudoidentity a such that $a + x(y + z) = xy + xz$.¹² This motivates us to define a weaker algebraic structure for the MI-(pre)field.

Definition 6.2. An algebra $(G, +, \cdot, E^0, E^1, f^0, f^1)$ is said to be a *weak MI-(pre)field*, if (F1) and (F2) hold and

(WF3) for any $x, y, z \in G$, there exists $a \in E^0$ such that the following *weak distributive law*

$$a + x(y + z) = xy + xz$$

holds.

Example 6.4. The sets of real intervals, fuzzy numbers, and parametric representations equipped with the arithmetical operations defined in Examples 4.1–4.4 form weak MI-fields. Note that M from Example 4.5 does not form a weak MI-field, since $M \setminus E^0$ also contains sets that are not multiplicative convex.

¹² We may express this equality as $x(y + z) \sim_+ xy + xz$, where $\sim_+$ denotes the congruence relations w.r.t. summation, see Proposition 4.3.

It should be noted that the distributivity law is very important for dealing with real numbers, e.g., mathematical formulas can be simplified or changed for a comparison. Thus, if one wants to use the benefit of the distributivity law for his/her work with fuzzy numbers, he/she must consider such the types of fuzzy numbers and arithmetics defined on them, which leads to an MI-field or to a similar but a bit weaker structure. A natural class of fuzzy numbers satisfying this assumption consists of the extensional fuzzy numbers, which are constructed using similarity relations, and their arithmetics do not stem from Zadeh's extensional principle (see [Example 6.1](#)). Unfortunately, satisfying the distributivity law may often result in the loss of the cancellation law. The weaker form of the MI-field (mainly the weak MI-field) seems to be an appropriate algebraic formalization of nearly all common approaches to the arithmetics of fuzzy numbers (intervals, convex compact sets, etc.). Thus, such a structure enables us to investigate the properties of such arithmetics in an abstract way and develop a deeper general theory. It is easy to see that by using the concepts introduced for the MI-groups, we may simply define analogous concepts for (weak) MI-fields, e.g., (canonical) MI-subfield and homomorphism of (weak) MI-fields. Nevertheless, such a development of MI-field theory is beyond the scope of this contribution and should be the subject of an individual paper with some theoretical results on the MI-fields. The same holds for the development of the MI-group theory.

7. Final remarks

7.1. Some reflections on “Computation over Fuzzy Quantities”

It has been a well-known fact [\[27\]](#) that if we consider a set of standard (LR, LU or most usual definitions) fuzzy numbers $\mathcal{F}(\mathbb{R})$ and the operations $+$, $\cdot$ defined with the help of Zadeh's extensional principle or α -cut based calculus, we obtain the following two monoids

$$(\mathcal{F}(\mathbb{R}), +, \tilde{0}) \quad \text{and} \quad (\mathcal{F}(\mathbb{R}), \cdot, \tilde{1}).$$

However, due to the singleton identity elements $\tilde{0}$ and $\tilde{1}$, we are not able to determine inverse elements, and thus we cannot proceed further towards a group structure or even towards a field structure. This illustrates the importance of introducing the MI-monoid, i.e., a structure that generalizes a monoid. Although it is not a problem to obtain a monoid using the previous approaches based on the extensional principle, the MI-monoid allows us to continue the development of more advanced structures.

Our work introduced in this paper is mainly motivated by the lack of a field-like structure for the so far existing arithmetics of fuzzy numbers. To the best of our knowledge, most of the articles focusing on the arithmetics of fuzzy numbers were rather application, technically or computationally oriented than investigating algebraic properties. However, there are interesting and very valuable exceptions. In our opinion, we may highlight one such exception that deserves special focus among the authors who devoted their research to the algebraic backgrounds of arithmetics of fuzzy numbers. It is the work of M. Mareš, which was published in many articles in the late 80s and early 90s and which is mostly presented in the book “*Computation over Fuzzy Quantities*” [\[27\]](#).

Obviously, although our and Mareš's approach to computation over fuzzy quantities stem from completely different roots and thus lead to different results, there is a non-trivial overlap between the two theories.

In [\[27\]](#), we learn that the author also finds the non-existence of inverse elements to be one of two crucial problems. The other one is – not surprisingly – the fact that the distributive law does not hold. As in our case, M. Mareš also states:

“it would be absurd to expect that arithmetic manipulation with vague fuzzy numbers will result in a crisp number, either $\mathbf{0}$ or $\mathbf{1}$ ”.

The use of the above idea in [\[27\]](#) is slightly different from our approach. The author first defines a so-called *y-symmetric fuzzy quantity* $a \in \mathcal{F}(\mathbb{R})$ as a fuzzy set for which the following holds:

$$a(y + x) = a(y - x), \quad x \in \mathbb{R} \tag{13}$$

and denotes the set of *y-symmetric fuzzy quantities* as $\mathbb{S}_y$. The set of symmetric fuzzy quantities is a union of all the *y-symmetric* ones over the whole real domain. Obviously, all the extensional fuzzy sets involved in our approach

are symmetric as well, which arises directly from the definition of the similarity relation. The important point in the approach provided in [27] is that the *additive equivalence* is defined. Two fuzzy quantities $a, b \in \mathcal{F}(\mathbb{R})$ are said to be *additively equivalent*¹³ (denoted by $a \sim_+ b$) if there exist 0-symmetric quantities $s_1, s_2 \in \mathbb{S}_0$ such that

$$a + s_1 \sim_+ b + s_2. \quad (14)$$

In other words, although the approach based on Zadeh's extensional principle (which is repeated in [27]) does enforce crisp identity elements, the author defines a sort of tolerance of given equalities up to some 0-symmetric fuzzy quantities. In our approach, these 0-symmetric fuzzy quantities would be elements of P_G , and consequently, they would be elements of E^0 , i.e., pseudoidentities.

The crucial difference lies in the fact that 0-symmetric elements from (14) may be arbitrarily wide. In our approach, which does not involve the extensional principle and suggests a completely different calculus, the width (vagueness) of pseudoidentity elements is given by the width (vagueness) of the original elements.

Note that, as our approach uses the $*$ -similarity relation as a crucial conceptual object, the same may be said about the approach by M. Mareš. The difference lies in the fact that we use it for the conceptual construction of extensional fuzzy numbers, i.e., for the construction of objects on which we develop the arithmetic. In the case of [27], the author used the similarity relation as a generalization of the classical equality to weaken the algebraic properties. Particularly, it can be shown that $\sim_+$ is a $*$ -similarity relation for $*$ being the Gödel t-norm. Thus, $a = b$ implies $a \sim_+ b$, and the monoid structure axioms are also valid in their weakened form

$$\begin{aligned} a + b &\sim_+ b + a, \\ a + (b + c) &\sim_+ (a + b) + c, \\ a + \mathbf{0} &\sim_+ a. \end{aligned}$$

Furthermore, in this weakened form, the other desirable properties also hold:

$$\begin{aligned} a + s &\sim_+ a, \quad s \in \mathbb{S}_0, \\ a + (-a) &\sim_+ \mathbf{0}, \end{aligned}$$

and thus a type of group-like structure is obtained.

Although the whole calculus is still based on Zadeh's extensional principle and is thus different from the one we suggested, the overlap of crucial ideas is obvious. No matter that our approach provides some unquestionable advantages such as computational simplicity, the limitation of undesirable widening of fuzzy numbers, preserving the required shape of fuzzy numbers, a field-like (MI-field) structure, i.e., the distributivity law, and an intuitive way of calculating with extensional fuzzy numbers, we are very grateful for the work by M. Mareš, which was mainly presented in [27] and followed subsequently, e.g., in [28]. It provided us strong motivation.

7.2. Conclusions

We have reviewed standard approaches to arithmetics of fuzzy numbers stemming from Zadeh's extensional principle and the drawbacks of such approaches. We have listed the crucial points for such drawbacks: that fuzzy numbers do not conceptually stem from numbers but from sets and that Zadeh's extensional principle enforces the fact that only the singletons $\tilde{0}, \tilde{1}$ may serve as the identity elements.

With these observations, and motivated by the importance of arithmetics of vaguely defined numbers, we have introduced a novel approach that attempts to mimic the arithmetic of real numbers from the very roots. This approach means that we have started from the conceptually clear definition of an extensional fuzzy number and introduced arithmetic operations on such objects that obey human intuition. Further, we have shown that with this approach, we obtain an algebraic structure that is isomorphic to the field of real numbers.

Motivated by this creation and by the natural requirement to generalize the structure for more similarity relations that model tolerance measures, we came up with a generalization of such an arithmetic for whole (even uncountable) systems of similarity relations. We have demonstrated very promising algebraic properties that are very similar to

¹³ Multiplicative equivalence is defined analogously.

the highly desirable properties of groups and fields with the only difference, the non-uniqueness of identity elements in such structures. Moreover, we have shown that if the system of similarities is appropriately chosen, e.g., as in the case of the embedded closed systems of shift-invariant similarities, the suggested arithmetic provides us with unquestionable practical advantages including the computational efficiency.

To formally describe an algebraic structure for the introduced framework of such arithmetics, we have presented novel many identities algebras, namely MI-monoids, MI-groups, and finally MI-fields. These MI-algebras capture the required properties due to a set of (pseudo)identity elements. In many examples, we have shown not only that MI-algebras formalize the suggested arithmetic of extensional fuzzy numbers but also that many well-known arithmetics of intervals, fuzzy numbers or sets of real vectors can be viewed from the MI-algebras perspective.

This fact enables us to investigate abstractly the properties of structures of various types of sets expressing imprecisely defined (not necessarily real) numbers. Moreover, we believe that the development of a theory of MI-algebraic structures should be interesting from a theoretical as well as a practical point of view. In this paper, we presented several results of this possible algebraic theory to demonstrate its functionality and a close relationship to the standard results from the theory of algebraic structures that serve as an algebraic backgrounds for the arithmetic of real numbers. It should be noted that the results for MI-algebras cannot be obtained by a trivial reconstruction of known proofs for algebras, but instead must be designed by novel consideration. Thus, the development of a theory of MI-algebras becomes a challenge for our future research.

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